MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers P5 Tutor Guide

Holiday program for the Primary 5 level

Notice it. Test it. Explain why.

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Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

The program at a glance

Level	Children whose school year covers Primary 5 topics (ages 10 to 12), going into P6, the PSLE year, who are ready for more thinking than standard practice
Length	10 lessons of 90 minutes, two a week over the 5 weeks of the holidays
Group size	Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8.
Focus	Thinking skills: making a list in order, guess and check, drawing a model, working backwards, finding patterns, reasoning and explaining. Not speed drills.
The arc of every lesson	Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why.
What children take home	A student workbook with an inquiry journal, 10 worksheets and a certificate
Before tutors start	Every tutor signs the tutor confidentiality agreement before receiving any materials.

Schedule

Week	Lesson	Title	Topic	Big idea
Week 1a	1	**Million Puzzles**	Numbers to 10 Million	Place value to 10 million
Week 1b	2	**Fraction Feats**	Fractions: Four Operations	Fraction and division (whole number ÷ whole number)

Week 2a	3	**Before and After**	Fraction Word Problems	Drawing bar models
Week 2b	4	**Triangle Areas**	Area of Triangle	Spotting the height
Week 3a	5	**Volume Vault**	Volume	Make a list in order
Week 3b	6	**Rate Racers**	Rate	Find the value of 1 unit
Week 4a	7	**Percentage Puzzles**	Percentage	Asking "percentage of what?"
Week 4b	8	**Angle Chasers**	Angles, Triangles and Quadrilaterals	Angle-chase chains with reasons
Week 5a	9	**Pattern Generalisers**	Patterns and Number Sense	Looking for a pattern
Week 5b	10	**Challenge Day**	Mixed Strategies	Choosing a strategy

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly.

Minutes	Part	What happens
0–3	**Sneak-peek puzzles**	Two puzzles on ideas we have not learnt yet. Answers shown privately. Nobody is marked.
3–9	**Warm-up**	A quick game, then the quick questions.
9–21	**Explore**	Hands-on in pairs. Objects first. A second what-if question is on the slide for anyone who wants more.
21–33	**Mini-lesson**	Worked examples with click-to-reveal steps, and your-turn questions.
33–37	**Brain break**	Stand, stretch, drink water. Optional Story break (2 minutes, skippable): a short true story about a mathematician linked to the lesson, with one optional try-it question.
37–49	**Workbook page 1 (pairs)**	Start here. After the choice question (question 3; question 2 in Lesson 1) you may choose: carry on, or try Think harder.
49–53	**Check page 1 together**	Four minutes, so every part can be explained.
53–58	**Short teaching before page 2**	One or two worked examples for the page 2 question types.
58–72	**Workbook page 2 (alone)**	A menu, not a wall: start with the questions named at the top, then choose. Nobody has to finish. Minutes 62 to 72 are protected open think time (10 quiet minutes) for Think harder, Go further or the Investigation.
72–76	**Check page 2 together**	Four minutes.
76–82	**Talk time**	Pairs talk, volunteers share, then everyone tries the first step of the Investigation. Pass is always allowed.
82–86	**Exit ticket**	One question on a slip of paper.
86–88	**Reflection (2 minutes)**	The small box on the Investigation and reflection page, including "Next time I will try". Pass is always allowed.
88–90	**Stamp and goodbye**	Stamp the inquiry journal. Lesson 10: certificates.

This is the one timeline used by the slides, the scripts and the workbook. Online, children keep cameras on and share work by holding paper to the camera or by showing it to the tutor only, on camera. Face to face, children sit in pairs and the tutor circulates.

Teaching habits that matter

- Ask “How do you know?” after every answer, right or wrong.
- Wait 10 seconds before helping. Let children struggle a little.

- Talk time is by invitation: ask for volunteers first, never cold-call a child by name, always allow “pass”, and never keep public tallies or rankings.
- Praise the thinking (“You checked it two ways”), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child finishes early, offer the Go further what-if or the Investigation, not extra drill.

Preparing the materials

- Send the workbook PDF to parents one week before. Ask them to print it single-sided and the cut-out pages on card if they can.
- Cut-outs used: square grid and dot grid (Lessons 4, 8), fraction wall (Lessons 2, 3), cube net (Lesson 5). Children need a protractor and a ruler.
- Keep the answer key with the tutor only. Check answers together on screen.
- Use the shared screen or a digital whiteboard for the mini-lesson table.

Setting up online

- Send parents the workbook and materials list one week before. Needed: the printed workbook and cut-outs, scissors, a ruler, paper and a pencil.
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child’s page and hands.
- Hold up the answer page to the camera at the end of each lesson, or ask the parent to take a photo and send it to the tutor through the approved parent channel (the child does not send it directly).
- Do not record lessons (this is in the tutor agreement). Add a second tutor or helper above 8 children.

Materials master list

Lesson	Title	Face to face	Online set-up
1	Million Puzzles	Digit cards 0 to 9 (one set per child), a 7-column place-value mat (M, HTh, TTh, Th, H, T, O), mini whiteboard, a calculator for the explore (every child needs one; a phone calculator is fine), workbook.	Children cut out paper digit cards and draw a 7-box place-value mat before the lesson and hold them up to the camera. The tutor shares the slides and writes on the place-value chart. Pairs work in breakout rooms for the explore and page 1. Before the lesson, ask every parent to put a calculator (a phone calculator is fine) next to the child for the explore.
2	Fraction Feats	Paper circles (6 per child) to cut as pizzas, scissors, a strip of paper for bar models, digit cards 1 to 4, mini whiteboard, workbook.	Children cut out six paper circles before the lesson and fold or cut them on camera. The tutor shares the slides and draws the bar models live. Pairs work in breakout rooms for the explore and page 1.
3	Before and After	Mini whiteboard and marker, 24 counters or paper squares (two colours), paper strips for bar models, workbook, pencil.	The tutor shares the slides and builds each bar model click by click. Children draw their models on whiteboards and hold them up to the camera together. For the explore, children move 18 + 6 paper squares on the desk with the camera up as usual and hold the table up to the camera; or use the interactive version on the slide.
4	Triangle Areas	Square grid paper (cut-out page at the back of the workbook), scissors, ruler, set square or the corner of a card, mini whiteboard and marker, workbook, pencil.	The tutor shares the slides and clicks to draw heights and reveal each step. For the explore, children draw and cut their rectangle with the camera up as usual, and hold the cut pieces up to the camera to show they fit (or use the interactive version on the slide). Whiteboards are held up together for the quick questions.

5	Volume Vault	About 30 one-centimetre cubes (or dice, Lego bricks, sugar cubes), a ruler, squared paper, mini whiteboards. Tutor: a clear plastic box with a ruler taped to its side, water and a stone for the demonstration.	The tutor shows the water rise live on camera: a clear box, a ruler taped to the side, a stone dropped in. Children use dice or Lego for the Explore; with no cubes they draw layers on squared paper. Whiteboards are held up to the camera.
6	Rate Racers	Mini whiteboards, a calculator for checking only, a stopwatch (or phone), a few real price tags or supermarket flyers if you have them.	Children hold whiteboards up to the camera; the tutor fills the rate tables on the slides. Pairs work in breakout rooms for the Explore. The brain break uses a phone stopwatch.
7	Percentage Puzzles	Mini whiteboard and marker, workbook, pencil, a hundred-square page (or squared paper), a calculator for checking only at the end of each page.	Children keep the workbook and a whiteboard in front of the camera. The tutor screen-shares the slides and draws bar models with the annotation tool. Calculators stay closed until the check-together slides.
8	Angle Chasers	Paper, ruler, pencil, scissors, protractor, red and blue pencils, the dot-grid cut-out, mini whiteboards, workbook.	Children tear or cut the corners of a paper triangle and quadrilateral on camera. The tutor screen-shares the slides and draws on the figures with the annotation tool. Children keep a red pencil for the angle they are chasing.
9	Pattern Generalisers	Matchsticks or toothpicks (about 20 per child, 40 if possible), square paper tiles or the square grid from the cut-out pages, mini whiteboard, workbook.	Children have toothpicks and paper squares ready on the desk and build each figure on camera. The tutor shares the slides; pairs work in breakout rooms for the explore and page 1, and hold their tables up to the camera.
10	Challenge Day	Four puzzle cards shown on the slide, or sent as images to each breakout room (backwards, logic, counting, guess and check), mini whiteboards, scrap paper, a 3 by 4 grid drawn on paper, certificates, stamps.	Each breakout pair gets one puzzle card on screen and reports back how they started; logic grids and models are shared on screen and annotated together. Children hold their working up to the camera.

How to use the scripts

Every lesson below is a minute-by-minute script. The slides match it: one slide or two for each part. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

SAY	Words to say aloud, in plain language for 10- to 12-year-olds. Said in your own voice, not read from the page.
DO	What you do: set up, hand out, show a slide, or move around the room.
ASK / EXPECT	Questions to put to the group, with the answer you are listening for. Let children answer before you explain.
IF STUCK	A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer.

Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too, so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

Sneak-peek puzzles, choosing a start and the reflection

- Every lesson starts with 3 minutes of sneak-peek puzzles: two puzzles on ideas we have not learnt yet. The first is an access puzzle any child can try with objects, drawing or counting, so everybody starts with a win. The second is a thinking puzzle. Say clearly: guessing and “I do not know” are welcome and nobody is marked.
- Answers are shown privately: on the child’s own paper held to you as you walk round. Online, everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Never use boards in front of the group, never read answers out with names, never keep a tally or a ranking. Ask for volunteers to say how they thought.
- Choosing a start: every child has the same neutral line on workbook page 1, “After question 3 you may choose: carry on, or try Think harder.” (In Lesson 1 the line says “After question 2”.) Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never announce who is on which page, never name a child in front of the group for this, and do not split a pair unless both children agree. Any child may swap at any time. The Go further what-if on the explore slide and the Investigation (printed on the page every child gets) give anyone more to do without waiting.
- Every lesson ends with a 2-minute reflection in a small box on the Investigation and reflection page: “Something I did well”, “The hardest part was…”, “What helped me” ticks that include “I asked for help” and “I am still stuck”, and “Next time I will try …”. Pass is always allowed. It is never collected or read out in public; you may read it to plan the next lesson. Follow-up for “I am still stuck” is defined: in the

NEXT lesson, in the open session, at minutes 37 to 49 (page 1 pair work), visit those children first in their shared breakout room with the camera on and a parent nearby (face to face: at their table), and go through the idea with one worked example. You may also re-teach the most-ticked idea to the whole group in the warm-up. After Lesson 10 there is no next lesson, so the follow-up goes to the parent through the approved parent channel. Praise strategy, not speed.

- Open think time: minutes 62 to 72 of every lesson are protected, 10 quiet minutes for any child who wants a longer problem (Think harder, Go further or the Investigation). No hints unless a child asks and no calling on anyone. Children still on Start here carry on with it. It is open to every child, never announced as a group for particular children.

Answering a wrong answer

If...	Then...
A child gives a wrong answer	“Thank you. Let us test it with the counters.” Then build it together.
A child gives a right answer but cannot explain	“Show me with your counters.” Then: “Tell your partner how you know.”
A child says “I do not know”	“What do you notice? What could you try first?” Offer hint 1.
Two children disagree	“Both of you, convince the group using objects.” Let the objects decide.

Level 5 mathematician stories

Every lesson has an optional 2-minute Story break inside the brain break (minutes 33 to 37): a short true story about one mathematician, shown on its own slide, with one optional try-it question. It is skippable: if you are short of time or the group needs the full break, say "we will skip the story today" and go on to the workbook at minute 37. Nobody is marked, children are invited as a whole group (volunteers only, pass is always fine), and the try-it answer is in the speaker notes and in the answer key for that lesson. Each story also appears as a small corner box on the Investigation page of the workbook (no answer there). The researched sources are in docs/stories/final_p5.json.

Lesson	Person	Country	Years	Link to the lesson
1	**Dorothy Vaughan**	United States	1910–2008	Today you add, subtract, multiply and divide numbers up to ten million, the same four operations that human computers did carefully and checked by hand.
2	**Eugenia Cheng**	England (now works in the United States)	born 1976	Today you add, subtract and share with fractions, just like sharing a cake or following a recipe.
3	**Alcuin of York**	England (Northumbria), later at the court of Charlemagne in Europe	about 735–804	Today you solve fraction word problems by drawing before-and-after models, and Alcuin's stork puzzle can be solved with a model like that.
4	**Liu Hui**	China (kingdom of Wei)	3rd century CE (dates uncertain, roughly 220s to 290s)	Today you find triangle areas, and Liu Hui shows that it helps to ask why a rule works, not just how to use it.
5	**Archimedes**	Syracuse, Sicily (ancient Greek world)	about 287–about 212 BC	Today you work out volumes of cuboids and water levels, and the bath story is about using the water a solid pushes aside to measure volume.
6	**Mary Jackson**	United States	1921–2005	A wind tunnel test is full of rates: how far air or a plane moves in each hour or each second, just like the rates you work with today.

7	**Adam Ries (also spelled Adam Riese)**	Germany	1492–1559	Today you work out discounts, GST and price changes with percentages, the kind of everyday money sums that Ries taught ordinary people to do.
8	**Maryam Mirzakhani**	Iran (later worked in the United States)	1977–2017	A bouncing ball's path is made of straight lines and angles, and today you find unknown angles on a straight line.
9	**Al-Karaji**	Baghdad (today Iraq); some sources say he was born in Karaj, Iran	about 953–1029	Today you look for patterns, say the rule in words and test it with more terms, which is exactly what Al-Karaji did with his cube pattern.
10	**George Polya**	Hungary (later Switzerland and the United States)	1887–1985	Today is Challenge Day, and Polya's ideas such as drawing a picture, working backwards and checking are the tools you choose from.

Telling the stories honestly

- Say the label when there is one. Famous tales are told as tales: "a story often told about...", "historians are not sure".
- Lesson 3 (Alcuin of York): Historians are almost, but not completely, certain that Alcuin collected the puzzle book; the stork puzzle in the try-it is our own modern wording.
- Lesson 5 (Archimedes): A story often told about Archimedes (the bath and the wreath) is from a writer about two hundred years later; historians doubt the details.
- Lesson 7 (Adam Ries): Sources give his birth year as 1492 or 1493. The sources I opened do not name 'percentages' specifically; they list prices, interest, and profit and loss.
- Lesson 9 (Al-Karaji): His birthplace and the origin of his name are debated by historians. Some of his work survives only through later writers.
- Do not add dialogue, feelings or childhood scenes that are not in the story. If a child asks something you do not know, say "good question, let us find out", not a guess.
- Lesson 6: the try-it uses rate (distance in each hour) only, because the program does not teach speed.

Lesson 1: Million Puzzles

Week 1 (first lesson) • Numbers to 10 Million • Workbook: Lesson 1

Objective	Read, write and compare numbers up to 10 million; solve digit-card and digit-riddle puzzles with place value (testing the best number just below and just above a target); estimate to check large calculations; and use the order of operations and brackets to hit target numbers.
Thinking skills	Place value to 10 million, testing both sides of a target, reasoning from clues in order, estimating by rounding, order of operations with brackets, making a list in order
Materials	Digit cards 0 to 9 (one set per child), a 7-column place-value mat (M, HTh, TTh, Th, H, T, O), mini whiteboard, a calculator for the explore (every child needs one; a phone calculator is fine), workbook.
Online set-up	Children cut out paper digit cards and draw a 7-box place-value mat before the lesson and hold them up to the camera. The tutor shares the slides and writes on the place-value chart. Pairs work in breakout rooms for the explore and page 1. Before the lesson, ask every parent to put a calculator (a phone calculator is fine) next to the child for the explore.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *"Welcome to Math Inquirers P5! This holiday we play with puzzles that stretch you before P6. Every lesson we notice a pattern, test it and explain why."*

"Today our numbers grow up to 10 million. How big is a million, really? By the end you will know, and you will solve puzzles that most adults get wrong."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): Use the digit cards 3, 6, 7 and 9, each once, to make a 4-digit number as close to 7000 as you can. Try one that starts with 7 and one that starts with 6. Which is closer?	Starting with 7, the best is 7369 (369 above 7000). Starting with 6, the best is 6973 (27 below 7000). 6973 is closer, and it is the closest of all 24 numbers. A child who only tries numbers that start with 7 misses it: testing BOTH sides of a target is today's idea. Any honest try is a good start.
Puzzle 2 (thinking): Put ONE pair of brackets in 4 + 6 × 5 − 3 to make 47. How do you know no other place makes 47?	(4 + 6) × 5 − 3 = 47. The other single pairs give 16 or 31 (and so does no bracket at all). Brackets first, then × and ÷, then + and −.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked."

Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 2 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Million Maker. Draw seven boxes (M to O). The tutor turns over digit cards one at a time; write each one in a box straight away and do not move it. Closest to 5 000 000 wins. Play 3 rounds.

SAY *"Draw seven boxes for millions down to ones. I turn over one digit card at a time. Write it in a box straight away. No moving it later!"*

"The winner is the number closest to 5 000 000. Think: where does a 4 go? A 9? A 0?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Write in numerals: 3 million, 40 thousand and 6	3 040 006
The value of the 7 in 2 716 450	700 000
4 560 000 ÷ 1000	4560
Round 6 849 500 to the nearest thousand	6 850 000
8 + 4 × 5 − 6 ÷ 2	25

9–21 min Explore (in pairs, with objects)

SAY *"Before any working: how long is a million seconds? An hour, a day, a week, a month or a year? Write your guess in the corner of your whiteboard and do not change it."*

DO 1. Guess first: is a million seconds about an hour, a day, a week, a month or a year? Write your guess.

2. Work out how many seconds there are in 1 hour.

3. Work out how many seconds there are in 1 day.

4. How many days is 1 000 000 seconds? Get as close as you can, then compare with your guess.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: how many seconds in one minute? In one hour? Multiply, do not add.

Stuck on days: you know the seconds in one day. How many days make close to 1 000 000? Try 10 days first.

Finished early: how long is 10 million seconds? More or less than a school term?

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): A million seconds is about 11 and a half days. What about a BILLION seconds (1 000 000 000)? Guess first: days, months or years? Then use your calculator. (Answer: About 11 574 days, which is about 32 years. 1 000 000 000 ÷ 86 400 = 11 574.07 days; 11 574 ÷ 365 is about 31.7 years. A billion is a thousand millions, so 1000 × 11.6 days.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many seconds are in 1 day?	86 400 (60 × 60 = 3600 in an hour, × 24).
About how many days is a million seconds?	Between 11 and 12 days: 11 days = 950 400 s and 12 days = 1 036 800 s. About 11 and a half days.
About how long is 10 million seconds?	About 116 days, nearly 4 months (10 × 11.6 days).

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Cards 0, 1, 2, 4, 5, 7, 8. I want the number closest to 5 000 000. I could start with 5 and make the smallest number after it: 5 012 478. That is 12 478 too big."*

"But I must also look just below the target. Start with 4 and make the biggest number after it: 4 875 210. That is 124 790 too small. I compare the two gaps: 12 478 is smaller, so 5 012 478 wins. I always test both sides."

"For a digit riddle I use the clue that fixes the most digits first. Rounded to the nearest hundred thousand it is 3 600 000, so it starts 3 5 with a big digit next, or 3 6 with a small digit next. Then I follow the other clues one by one and check every clue at the end."

Worked example on the slides	Answer

Closest to 5 000 000 with the cards 0, 1, 2, 4, 5, 7, 8. Just above: start with 5, then the smallest: 5 012 478 (12 478 away). Just below: start with 4, then the biggest: 4 875 210 (124 790 away). Compare the gaps.	5 012 478 is closest. Always test both sides of the target.
Digit riddle: rounds to 3 600 000 (nearest hundred thousand); ten thousands digit is 4 more than the hundred thousands digit; thousands digit is their difference; hundreds digit is half the thousands digit; tens digit 0; digits add to 31; all different. Rounding: 3 5 with a big digit next, or 3 6 with a small digit next. 4 more: 5 + 4 = 9 works; 6 + 4 = 10 does not. Thousands 9 − 5 = 4, hundreds 2, tens 0, ones 31 − 23 = 8.	3 594 208. Check every clue at the end.

SAY *"The key idea today is: test both sides, check every clue. Place value tells you what each digit is worth. To get closest to a target, test the best number just below AND just above. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Leaving out zeros that hold a place, for example writing three million, forty thousand and six as 340 006 instead of 3 040 006, or mixing up the value of a digit (3000) with how many thousands a number has (4053).
- Looking only on one side of the target: for "closest to 6 000 000" trying only numbers that start with 6 and missing 5 986 320, which is closer.
- Working from left to right and ignoring the order of operations (6 + 4 × 5 read as 50 instead of 26).

33–37 min Brain break

DO Human number line: stand up and stretch as tall as 10 million, crouch as small as 1, then show where 5 million, 2 500 000 and 9 900 000 would be with your arms.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Dorothy Vaughan, 1910–2008, United States." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today you add, subtract, multiply and divide numbers up to ten million, the same four operations that human computers did carefully and checked by hand. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): 2 450 000 + 1 375 000 = 3 825 000. 2 450 000 − 1 375 000 = 1 075 000. 1250 × 400 = 500 000. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say “Pairs first. Talk before you write.”

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

“After question 2 you may choose: carry on, or try Think harder. You can swap at any time.”

DO Circulate and ask:

- Ask: “Did you look just below AND just above the target?”
- Ask: “Which clue fixes the most digits? Start there.”
- Ask: “Estimate first. Is the calculator answer about the right size?”

IF STUCK Hint 1: “Read the number in groups of three from the right: millions, thousands, ones.”

Hint 2: “Write the two best numbers, one just below and one just above, and find each gap.”

Hint 3: “Put each clue’s digit straight into the place-value chart as you go.”

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Million Puzzles (What if we change the target?)	First step for everyone: Easy first step: use the cards 0, 1, 2, 4, 5, 7, 8 for just ONE target, 4 000 000. Write one number from just below it and one from just above it. Which is closer? Then do part (a). (a) Use the digit cards 0, 1, 2, 4, 5, 7, 8, each once, to make the 7-digit number closest to a TARGET. Find the closest number for the targets 3 000 000, 4 000 000, 5 000 000 and 6 000 000. (b) For each target, does the winner come from just BELOW the target or just ABOVE it? What do you notice? Try 7 000 000 and 8 000 000 to test your idea. (c) What stays the same in every case? Write a rule that tells someone which side wins.

What to look for / model answer	Closest: 3 000 000 gives 2 875 410 (below, 124 590 away); 4 000 000 gives 4 012 578 (above, 12 578); 5 000 000 gives 5 012 478 (above, 12 478); 6 000 000 gives 5 874 210 (below, 125 790); 7 000 000 gives 7 012 458 (above); 8 000 000 gives 8 012 457 (above). Rule: when the leading digit of the target (4, 5, 7, 8) is one of the cards, start with it and make the rest as small as possible, so the winner is just above. When the card is missing (3 and 6) the winner has to come from below, and the number just below starts with the next smaller card and is as big as possible. Look for: both sides tested every time, gaps compared by subtracting, a rule in words.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: How many 7-digit numbers have digits that add up to 2? List them all.	7 numbers: 1 000 001, 1 000 010, 1 000 100, 1 001 000, 1 010 000, 1 100 000 and 2 000 000.
Workbook Think harder 2: Put ONE pair of brackets into 2 + 3 × 4 + 5 to make 25. Then put one pair in a different place to make 29.	(2 + 3) × 4 + 5 = 25 and 2 + 3 × (4 + 5) = 29. (29 is the biggest you can make with one pair.)
Workbook Think harder 3: A whole number rounded to the nearest thousand is 640 000. What is the greatest it could be? The smallest?	Greatest 640 499, smallest 639 500.
Workbook Think harder 4: Use each of the digits 1, 2, 3, 4, 5, 6 and 7 once. What is the difference between the greatest and the smallest 7-digit number you can make?	6 419 754 (7 654 321 − 1 234 567).
Workbook Go further 5: Put brackets into 1 + 2 × 3 + 4 × 5 + 6 to make the answer as big as possible. You may use more than one pair. What is the biggest answer?	231: (1 + 2) × (3 + 4) × (5 + 6) = 3 × 7 × 11. With no brackets it is only 33.
Workbook Go further 6: I am a 7-digit odd number. My digits are all different and add up to 42. Rounded to the nearest hundred thousand, I am 7 000 000. I am the smallest number that fits. Who am I?	6 953 487. Seven different digits adding to 42 must be 3, 4, 5, 6, 7, 8, 9 (the biggest seven digits add to exactly 42), so there is no 0. The number is between 6 950 000 and 7 049 999, so it starts 6 9 with 5, 7 or 8 next; the smallest is 6 95 then 3, 4, 8, 7 (it must end in an odd digit).
Workbook Go further 7: Ali writes all the whole numbers from 1 to 1000. How many digits does he write? (Go further: how many if he writes all the numbers from 1 to 1 000 000?)	2893. 9 × 1 + 90 × 2 + 900 × 3 + 4 = 9 + 180 + 2700 + 4. Go further: 5 888 896 (9 × 1 + 90 × 2 + 900 × 3 + 9000 × 4 + 90 000 × 5 + 900 000 × 6 + 7 = 9 + 180 + 2700 + 36 000 + 450 000 + 5 400 000 + 7).

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Siti says a 7-digit number is always bigger than a 6-digit number. Ravi says not if the 6-digit number has bigger digits, like 999 999. Who is right?"*

"How did you check that you had found ALL the numbers in Q6?"

Discussion question	Listen for
Siti says a 7-digit number is always bigger than a 6-digit number. Ravi says not if the 6-digit number has bigger digits, like 999 999. Who is right? Explain.	Siti is right. The smallest 7-digit number is 1 000 000 and the biggest 6-digit number is 999 999, which is 1 less. A 7-digit number has a millions digit that is at least 1, and that alone beats every 6-digit number, whatever its digits are.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
(a) What is the value of the digit 4 in 8 406 125? (b) Add one pair of brackets to make it true: 5 + 3 × 8 – 2 = 62	(a) 400 000 (b) (5 + 3) × 8 – 2 = 62

SAY *"Today you read numbers up to 10 million, tested both sides of a target, followed clues to a single answer and used estimation and brackets. That is how inquirers work with big numbers."*

"Do the exit ticket on your own, then show me."

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach

the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "You got unstuck by drawing it. That is a real mathematician move."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

Use the digit cards 0 to 9, each at most once, to make the 7-digit number closest to 5 000 000. (5 012 346 is 12 346 away and 4 987 653 is 12 347 away, so 5 012 346 wins by 1.)

Answer key (workbook)

Q1 (a) 5 308 040 (b) 400 000; 8 (c) 3250

Q2 (a) 9 865 320 (b) 2 035 698 (c) Just below: 5 986 320 (13 680 away). Just above: 6 023 589 (23 589 away). Closest: 5 986 320.

Q3 7 253 609. Rounding gives 7 2 with ten thousands digit 5 or more, or 7 3 with ten thousands digit 4 or less. "3 more" only works for 7 25: 2 + 3 = 5. Then thousands 5 − 2 = 3, hundreds 6, tens 0, and 7 + 2 + 5 + 3 + 6 + 0 = 23, so the ones digit is 32 − 23 = 9.

Q4 (a) about 100 000: could be right (b) about 60 000: must be wrong (c) about 6 300 000: must be wrong (d) about 320 000: could be right

Q5 (a) (6 + 4) × 5 − 2 = 48 (b) 6 + 4 × (5 − 2) = 18 (c) 1 000 000 ÷ (200 + 300) × 2 = 4000 (d) 5 × (3 × 9 − 7) = 100

Q6 (a) 2 499 500 and 2 500 499 (b) 2 500 002, 2 500 011, 2 500 020, 2 500 101, 2 500 110, 2 500 200: 6 numbers. None below 2 500 000: 2 499 5__ already has digits adding to at least 29.

Story try-it (optional, Dorothy Vaughan) 2 450 000 + 1 375 000 = 3 825 000. 2 450 000 − 1 375 000 = 1 075 000. 1250 × 400 = 500 000.

Lesson 2: Fraction Feats

Week 1 (second lesson) • Fractions: Four Operations • Workbook: Lesson 2

Objective	Link fractions and division (3 ÷ 4 = 3/4); write fractions as the sum of two different unit fractions by testing in order; solve fraction-of-a-remainder problems forwards and backwards with a bar model; and spot and explain patterns in fraction sums.
Thinking skills	Fraction and division (whole number ÷ whole number), drawing a model (bar, pizza), testing in order and knowing when to stop, fraction of a remainder, working backwards, looking for a pattern, testing both ways
Materials	Paper circles (6 per child) to cut as pizzas, scissors, a strip of paper for bar models, digit cards 1 to 4, mini whiteboard, workbook.
Online set-up	Children cut out six paper circles before the lesson and fold or cut them on camera. The tutor shares the slides and draws the bar models live. Pairs work in breakout rooms for the explore and page 1.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *"Welcome back, inquirers! Today is pizza day: we share pizzas fairly, cut them in clever ways and find out what the Ancient Egyptians knew about fractions."*

"Notice it, test it, explain why: by the end you will solve fraction puzzles that trip up grown-ups."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): Two pizzas are shared equally among 4 friends. Draw the pizzas and show how you would cut them. How much pizza does each friend get?	1/2 of a pizza each. For example cut each pizza in half: that makes 4 halves, one for each friend. Any fair cutting is fine (for example cutting each pizza into quarters and giving each friend 2 quarters).
Puzzle 2 (thinking): Mei spends 1/3 of her money on a book. Then she spends 1/2 of the money she has LEFT on lunch. What fraction of her money has she spent altogether? Use the bar.	2/3. After the book 2/3 is left; half of 2/3 is 1/3 (lunch); 1/3 + 1/3 = 2/3. The half is a half of what is LEFT, not of everything.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked." Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Fraction Face-off. Two fractions appear. Hold up 1 finger if the first is bigger, 2 if the second is. Whisper a reason to your partner (same top? close to 1? more than a whole?); volunteers share. 3/5 or 3/7, 5/6 or 6/7, 4/9 or 1/2, 7/8 or 8/7.

SAY *"Fraction Face-off! Two fractions appear. One finger if the first is bigger, two fingers if the second is bigger."*

"Tell your partner a reason in a whisper: same top? Close to 1? More than a whole? Try it without common denominators first, but any method is welcome. I will only ask volunteers to share, and you may pass."

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
3 ÷ 4 as a fraction	3/4
2/3 + 1/6	5/6
3/4 of 20	15
Which is bigger, 1/2 or 3/5?	3/5
1/4 of 12	3

9–21 min Explore (in pairs, with objects)

SAY *"3 pizzas are shared equally among 4 friends. Will each friend get more than half a pizza or less? Write your prediction before you cut anything."*

DO 1. Take 3 paper pizzas. Share them equally among 4 friends.

2. How much does each friend get? Write it as a fraction.

3. Can you cut so that each friend gets only 2 pieces?

4. Now share 5 pizzas among 6 friends, with only 2 pieces each.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: cut every pizza into 4 equal pieces. How many pieces are there? How many does each friend get?

Too many pieces: can you cut some pizzas in halves and only one into quarters?

Finished early: share 5 pizzas among 6 friends so that each friend gets just 2 pieces.

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): 4 pizzas are shared among 5 children, 5 pizzas among 6 children, and 6 pizzas among 7 children. Who gets the biggest share? What do you notice as the group gets bigger? (Answer: 4/5, 5/6 and 6/7: the share grows each time (0.8, 0.83, 0.86). Each child is 1/5, 1/6, 1/7 short of a whole pizza, a smaller piece each time, so the shares creep up towards 1 pizza but never reach it.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How much pizza does each of the 4 friends get?	3/4 of a pizza (3 ÷ 4 = 3/4).
How can each friend get only 2 pieces?	Cut 2 pizzas into halves and 1 pizza into quarters: each friend gets 1/2 + 1/4 = 3/4.
5 pizzas among 6 friends, 2 pieces each?	Cut 3 pizzas into halves and 2 pizzas into thirds: each gets 1/2 + 1/3 = 5/6.

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"3 pizzas among 4 friends. If I cut each pizza into quarters, there are 12 quarters and each friend gets 3 of them: 3/4. So 3 ÷ 4 = 3/4. The top number is what is shared, the bottom number is how many share it."*

"Now 7/12 as two different unit fractions. I try in order. 1/2: 7/12 – 6/12 = 1/12, a unit fraction, yes. 1/3: 7/12 – 4/12 = 3/12 = 1/4, yes. 1/4: what is left is 1/3, the same pair again. Once my first part is smaller than half of 7/12, the other part would be bigger, so I stop. Two ways."

"Mei had 180 beads. She used 1/3 for a bracelet and 3/4 of the REMAINDER for a necklace. I draw a bar in 3 parts. The remainder is 2 parts, which I cut into 4 small units, so the whole bar is 6 small units. Bracelet 2, necklace 3, left 1. 6 units = 180, so 1 unit = 30. She has 30 beads left."

Worked example on the slides	Answer

Write 7/12 as the sum of two different unit fractions. Try 1/2: 7/12 − 6/12 = 1/12. Yes. Try 1/3: 7/12 − 4/12 = 3/12 = 1/4. Yes. Try 1/4: we get 1/3, the same pair. Stop: the first part is now less than half of 7/12.	Two ways: 1/2 + 1/12 and 1/3 + 1/4.
Mei had 180 beads. She used 1/3 of them for a bracelet and 3/4 of the remainder for a necklace. How many beads were left?. Bar in 3 parts: bracelet 1 part, remainder 2 parts. Cut the remainder into 4 small units: the whole bar is 6 units. Bracelet 2 units, necklace 3 units, left 1 unit. 6 units = 180.	1 unit = 30. 30 beads were left.

SAY *"The key idea today is: the fraction of WHAT?. Find the remainder first, then take the fraction of the remainder. Draw a model with equal small units. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Taking the second fraction of the whole amount instead of the remainder (2/3 of 240 instead of 2/3 of 180).
- Adding tops and bottoms: 1/2 + 1/3 = 2/5, or using the same unit fraction twice when the question asks for different ones.
- Writing 3 ÷ 4 as 4/3, or writing 7 ÷ 3 as 3/7.

33–37 min Brain break

DO Fraction freeze: walk on the spot. When the tutor calls a fraction, freeze in a pose: arms up for more than 1/2, arms out for exactly 1/2, crouch for less than 1/2.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Eugenia Cheng, born 1976, England and the United States." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today you add, subtract and share with fractions, just like sharing a cake or following a recipe. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): 1/3 + 1/4 = 4/12 + 3/12 = 7/12 eaten. 1 − 7/12 = 5/12 is left. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say "Pairs first. Talk before you write."

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has

to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

"After question 3 you may choose: carry on, or try Think harder. You can swap at any time."

DO Circulate and ask:

- Ask: "Is that a fraction of the whole or of the remainder?"
- Ask: "How do you know you have found all the ways? When can you stop?"
- Ask: "Show me your model. Are all the small units the same size?"

IF STUCK Hint 1: "Draw a bar. Cut it into the parts the first fraction tells you."

Hint 2: "Cut the remainder again so every unit in the bar is the same size."

Hint 3: "Take away the unit fraction and simplify. Is the top number 1?"

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Fraction Feats (Is it always true? Test and explain)	First step for everyone: Easy first step: cut a bar into 6 equal parts. 1/2 is ___ parts and 1/3 is ___ parts, so 1/2 − 1/3 = ___/6. Now try part (a). (a) Take two unit fractions that are next to each other, such as 1/3 and 1/4. Take the smaller away from the bigger: 1/3 − 1/4 = _____. Try at least four more pairs (1/4 and 1/5, 1/5 and 1/6, 1/9 and 1/10, 1/20 and 1/21). (b) What do you notice about the answer each time? Write your idea in words. Is it always true? Explain with a drawing (hint: cut a bar into 3 × 4 = 12 equal parts). (c) Use your idea to write 1/6 + 1/12 + 1/20 as one fraction without finding a common denominator. Then make up your own sum of this kind (three or four fractions) for a partner to solve.

What to look for / model answer	Each answer is a unit fraction whose bottom number is the two bottom numbers multiplied: 1/3 − 1/4 = 1/12 (3 × 4), 1/4 − 1/5 = 1/20, 1/5 − 1/6 = 1/30, 1/9 − 1/10 = 1/90, 1/20 − 1/21 = 1/420. It is always true for neighbouring unit fractions (checked for every pair from 1/1 and 1/2 up to 1/1999 and 1/2000): cut a bar into n × (n + 1) parts; 1/n is (n + 1) parts and 1/(n + 1) is n parts, so the difference is 1 part. Part 3: 1/6 = 1/2 − 1/3, 1/12 = 1/3 − 1/4, 1/20 = 1/4 − 1/5, so the sum is 1/2 − 1/5 = 3/10 (check: 10/60 + 5/60 + 3/60 = 18/60 = 3/10). Make-up-your-own: any chain of neighbouring-pair fractions works, for example 1/2 + 1/6 + 1/12 + 1/20 = 1 − 1/5 = 4/5; check that the partner's answer matches the one found by adding the fractions the long way. Accept any clear picture or well-chosen examples; watch for "it works for my five pairs so always" and ask for the reason.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Can 4/5 be written as the sum of two DIFFERENT unit fractions? Test in order and explain how you know when to stop.	No. The bigger unit fraction must be more than half of 4/5 (more than 2/5) and less than 4/5, so only 1/2 can be tried: 4/5 − 1/2 = 3/10, which is not a unit fraction. 1/3 is already less than 2/5, so we stop.
Workbook Think harder 2: Ben ate 1/3 of a pizza. Ali ate 1/4 of what was left. What fraction of the pizza is left?	1/2. 2/3 was left after Ben; Ali ate 1/4 of it, so 3/4 of 2/3 = 1/2 is left.
Workbook Think harder 3: Siti shares 3 cakes equally among 8 children. Ben shares 2 cakes equally among 5 children. Whose child gets more cake, and how much more?	Ben's children get more, by 1/40 of a cake. Siti: 3/8 = 15/40 of a cake each. Ben: 2/5 = 16/40 of a cake each. 16/40 − 15/40 = 1/40.
Workbook Think harder 4: Mei walked 3/5 of a path in the morning and 1/4 of the WHOLE path in the afternoon. She still had 600 m to walk. How long is the path?	4000 m. 3/5 + 1/4 = 17/20 of the path is done, so 3/20 is left. 3/20 is 600 m, so 1/20 is 200 m and the whole path is 20 × 200 = 4000 m.
Workbook Go further 5: Work out 1/6 + 1/12 + 1/20 + 1/30 + 1/42 without finding a common denominator for all five. (Hint: 1/6 = 1/2 − 1/3.)	5/14. Each fraction is a difference: (1/2 − 1/3) + (1/3 − 1/4) + (1/4 − 1/5) + (1/5 − 1/6) + (1/6 − 1/7). The middle parts cancel, leaving 1/2 − 1/7 = 5/14.
Workbook Go further 6: 5 pizzas are shared equally among 12 children. Each child must get exactly 2 pieces, and each piece must be a unit fraction of a pizza (like 1/3 or 1/4). How can the pizzas be cut?	Each child gets 5/12. Two ways: 1/3 + 1/12 (cut 4 pizzas into thirds and 1 pizza into twelfths), or 1/4 + 1/6 (cut 3 pizzas into quarters and 2 pizzas into sixths).

Workbook Go further 7: A tank is full of water. Every hour, 1/3 of the water that is in the tank is let out. After 3 hours there are 16 litres left. (a) How much water was in the full tank? (b) After how many hours is less than half of the original water left? (c) How much water has been let out by then?	(a) 54 litres: each hour 2/3 is left, 54 → 36 → 24 → 16. (b) 2 hours: 24 litres is less than half of 54 (27), but after 1 hour 36 litres is more than half. (c) 54 − 24 = 30 litres.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Ben says 1/2 + 1/3 = 2/5 because you add the tops and add the bottoms. Is he right? Use a pizza to show why."*

"In Q4, why is each sum 1 take away a unit fraction?"

Discussion question	Listen for
Ben says 1/2 + 1/3 = 2/5 because you add the tops and add the bottoms. Use a pizza picture to show why he is wrong.	He is wrong. 1/2 of a pizza is already more than 2/5, so adding 1/3 more cannot give 2/5. Cut into sixths: 1/2 = 3/6 and 1/3 = 2/6, so the total is 5/6. We can only add the tops when the pieces are the same size.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
(a) 4 pizzas are shared equally among 5 children. How much does each child get? (b) Ali had 60 marbles. He gave 1/3 of them to Mei and 1/2 of the remainder to Ben. How many marbles did Ali have left?	(a) 4/5 of a pizza (b) 20 marbles (60 − 20 = 40 left, then half of 40 is given away)

SAY *"Today you shared pizzas, found unit fractions by testing in order, took fractions of a remainder with a model and found a pattern in fraction sums. Great inquiring!"*

"Do the exit ticket on your own, then show me."

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I

asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "Drawing the bar again when it did not work is exactly right."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

Write 1 as the sum of three different unit fractions. Is there more than one way? (Only one: 1/2 + 1/3 + 1/6.)

Answer key (workbook)

Q1 (a) 5/8 (b) 9/4 = 2 1/4 m (c) Group A. 2/3 = 10/15 and 3/5 = 9/15. (Or: each child in A is 1/3 short of a whole pizza, in B 2/5 short, and 2/5 is more than 1/3.)

Q2 (a) 1/2 + 1/5 (b) 1/4 + 1/5 (c) 1/3 + 1/24 and 1/4 + 1/8. (1/5 leaves 7/40, not a unit fraction; after that the first part is less than half of 3/8, so stop.)

Q3 (a) 120 (b) 60 (c) 1/4. Ben 60, remainder 180, Ali 2/3 of 180 = 120, left 60 = 1/4 of 240.

Q4 (a) 1/30, 1/42. The bottom numbers are 1 × 2, 2 × 3, 3 × 4, 4 × 5, ... (b) 2/3, 3/4, 4/5 (c) 9/10. 1/90 is the 9th term (9 × 10), and each sum is 1 take away the next unit fraction.

Q5 (a) $120 (b) $36. Model: 10 equal units; game 4, book 3, left 3 units = $36, so 1 unit = $12.

Q6 (a) 4/1 + 3/2 = 5 1/2 (b) 1/3 + 2/4 = 5/6 (1/4 + 2/3 = 11/12 is bigger, so test both ways) (c) 1/4 + 2/3 = 11/12, only 1/12 away from 1

Story try-it (optional, Eugenia Cheng) 1/3 + 1/4 = 4/12 + 3/12 = 7/12 eaten. 1 − 7/12 = 5/12 is left.

Lesson 3: Before and After

Week 2 (first lesson) • Fraction Word Problems • Workbook: Lesson 3

Objective	Solve fraction word problems with units and before-and-after models: fraction of a remainder, giving between two people (the total stays the same), both getting or spending the same amount (the difference stays the same), and one amount that does not change.
Thinking skills	Drawing bar models, choosing what stays the same, splitting parts into equal units, working backwards, checking an answer against every condition
Materials	Mini whiteboard and marker, 24 counters or paper squares (two colours), paper strips for bar models, workbook, pencil.
Online set-up	The tutor shares the slides and builds each bar model click by click. Children draw their models on whiteboards and hold them up to the camera together. For the explore, children move 18 + 6 paper squares on the desk with the camera up as usual and hold the table up to the camera; or use the interactive version on the slide.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *"Welcome back, inquirers! Today's problems all have a story with a BEFORE and an AFTER: someone spends money, gives away stickers, or gets a bit older."*

"The trick that makes them easy is to notice what stays the same. Notice it, test it, explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): Ben is 8 and his father is 32. (a) How old is Ben's father when Ben is 12? (b) How much older than Ben is his father now? How much older when Ben is 12?	(a) 36. (b) 24 years older now, and 24 years older when Ben is 12: the gap does not change.
Puzzle 2 (thinking): Ali has 40 cards and Mei has 20. Ali gives Mei some cards and then they have the same number. How many cards did Ali give? What stays the same?	10. The TOTAL stays 60, so each has 30 afterwards and Ali gives 40 − 30 = 10. Counters or a drawing help.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked." Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO What's left? A jar has 120 marbles. Turn over one card at a time: give away 1/4 of the jar; give away 1/3 of what is left. Children draw what is left as a new bar and write how many are left after each card. (Tutor extension: a third card, 5/6 of what is left.)

SAY *"A jar has 120 marbles. I will turn over two cards. After each card, draw what is LEFT as a new bar and write how many marbles are left. Each card tells you what to take the fraction of: "of the jar" or "of what is left". I will read it twice."*

"Volunteers: how did you find 1/3 of what was left? Show your bar if you like."

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
What is 2/5 of 45?	18
What is 1/4 of 36?	9
What is half of 1/2?	1/4
Ali has 12 more stickers than Ben. Each gets 5 more. What is the difference now?	Still 12
Ali has 30 cards and Ben has 20. Ali gives Ben 5. What is the total now?	Still 50

9–21 min Explore (in pairs, with objects)

SAY *"Ali has 18 counters and Mei has 6. Ali gives Mei counters one at a time. Before you move any, predict: what will happen to the total? To the difference?"*

"Now both children get 2 more counters each time. Predict again: total or difference, which one stays the same?"

DO 1. Ali takes 18 counters. Mei takes 6 counters.

2. Ali gives Mei 1 counter at a time. After each move write Ali, Mei, the total and the difference in a table.

3. Start again from 18 and 6. This time give BOTH children 2 more counters each time. Fill in a new table.

4. What never changes in each table? When does Mei have 1/2 as many counters as Ali?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: make a table with four columns, Ali, Mei, total, difference, and fill one row after every move.

Not sure about the difference: line the two rows of counters up side by side and count the extra ones.

Finished early: how many counters must Ali give so that Mei has 1/3 as many as Ali? (He gives 0: she already does, 6 is 1/3 of 18.) What about 3/5 as many? (Ali gives 3: 15 and 9.)

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): Ali has 19 counters and Mei has 6 (25 altogether). Ali gives Mei counters one at a time. Can Mei ever have exactly as many as Ali? Why or why not? Try 24 altogether too. (Answer: No for 25: equal shares would be 12 and a half each, so an odd total can never be shared equally. With 24 (18 and 6) it works: 12 each after Ali gives 6. The total never changes when they only swap counters.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
What stayed the same when Ali gave counters to Mei?	The total: 24 every time.
What stayed the same when both children got 2 more?	The difference: 12 every time.
When did Mei have 1/2 as many as Ali?	After Ali gave 2 (16 and 8), and after both got 6 more (24 and 12).

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Mei spent 2/5 on a bag, so I draw 5 parts and shade 2. Then 1/2 of the REMAINDER on shoes. The remainder is 3 parts and I can't halve 3 parts nicely, so I cut every part into 2. Now there are 10 equal units: bag 4, shoes 3, left 3."*

"3 units are $27, so 1 unit is $9 and 10 units are $90. Check: bag $36, shoes $27, left $27. That adds to $90."

"Ali gives Ben 14 cards. Did the total change? No, the cards only moved. So the total after is still 96. After, Ali has 1/3 as many as Ben: Ali 1 unit, Ben 3 units. 4 units = 96, 1 unit = 24. Ali had 14 more before: 38."

Worked example on the slides	Answer
Fraction of a remainder. Mei spent 2/5 of her money on a bag and 1/2 of the remainder on shoes. She had $27 left. How much money did she have at first? Cut the remaining parts into equal units.	$90 (10 units; 3 units = $27, 1 unit = $9).
The total stays the same. Ali and Ben had 96 cards altogether. Ali gave Ben 14 cards. Then Ali had 1/3 as many cards as Ben. How many cards did Ali have at first?	38 (after: 4 units = 96, Ali 1 unit = 24; 24 + 14 = 38).

SAY *"The key idea today is: find what stays the same. Ask what stays the same: the total, the difference, or one amount. Draw it the same in the Before and the After model. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Taking the second fraction of the whole amount instead of the remainder (1/4 of the remainder read as 1/4 of all the money).
- Thinking the difference goes down by 10 when one person gives the other 10 (it goes down by 20: one loses 10, the other gains 10).
- Keeping the units from the Before model when the thing that stays the same is a different amount, so the units in the two models are not the same size.

33–37 min Brain break

DO Fraction stretch. Arms wide is the whole. Show 1/2, then 1/4, then 3/4 of the whole with your arms. Touch your toes for 0!

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Alcuin of York, about 735–804, England and Charlemagne's court." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today you solve fraction word problems by drawing before-and-after models, and Alcuin's stork puzzle can be solved with a model like that. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): 3 units + 1/2 unit = 3 1/2 units = 100 − 2 = 98. 1 unit = 98 ÷ 3 1/2 = 28 storks. Check: 3 × 28 = 84, half of 28 = 14, and 84 + 14 + 2 = 100. Keep the label: Historians are almost, but not completely, certain that Alcuin collected the puzzle book; the stork puzzle in the try-it is our own modern wording. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say “Pairs first. Talk before you write.”

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

“After question 3 you may choose: carry on, or try Think harder. You can swap at any time.”

DO Circulate and ask:

- Ask: “What stays the same in this story: the total, the difference, or one amount? How do you know?”
- Ask: “Are your units all the same size? Point to one unit and tell me what it is worth.”
- Ask: “Check your answer in the story. Does every sentence still work?”

IF STUCK Hint 1: “Draw the AFTER model first. That is the part with the fraction.”

Hint 2: “Which amount did not change? Label it with its number in BOTH models.”

Hint 3: “Find what 1 unit is worth, then work backwards to the start.”

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Before and After (What if we change the total?)	First step for everyone: Easy first step: the total is 48 + 12 = 60. Draw one bar for 60 and cut it into 3 equal parts. How many stickers is one part? Then do part (a). (a) Ali has 48 stickers and Mei has 12. Ali gives Mei some stickers. Afterwards Ali has exactly twice as many as Mei. How many does Ali give? Draw the After bars. (b) What if the total is different? Try the totals 45, 50, 30 and 60 in any way you like (Ali has more than Mei to start). For which totals can Ali end up with exactly TWICE as many as Mei, with whole stickers? For which can it NOT be done? (c) What stays the same? Write a rule that tells you which totals work, and explain with a bar.

What to look for / model answer	Part 1: total 60, After is 3 units (Ali 2 units, Mei 1 unit), 1 unit = 20, so Ali ends with 40 and Mei 20: Ali gives 8. Part 2: totals 45 (Ali 30, Mei 15), 30 (Ali 20, Mei 10) and 60 work; 50 does not (50 ÷ 3 is not a whole number). Rule: the total never changes when one child gives to the other, and the After has 3 equal units, so the total must be a multiple of 3 (checked for every total from 1 to 100). The amount given then depends on the start: it is (Ali's start) − 2/3 of the total. Look for: the total drawn as one bar split in 3, the "multiple of 3" idea, and children testing a total like 50 and explaining why it fails.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Ravi spent 1/4 of his money on Monday and 2/5 of the remainder on Tuesday. He spent $66 altogether. How much money did he have at first?	$120 (20 units: Monday 5 units, Tuesday 6 units; 11 units = $66, so 1 unit = $6)
Workbook Think harder 2: A box had red pens and blue pens. 2/3 of the pens were blue. After 12 blue pens were taken out, 1/2 of the pens were blue. How many pens were in the box at first?	36 (the red pens did not change: red 1 unit, blue 2 units at first, 1 unit after; 1 unit = 12)
Workbook Think harder 3: Mrs Tan is 41 years old and her son is 9. How many years ago was her son's age 1/5 of her age?	1 year ago (the difference, 32, is 4 units; 1 unit = 8: son 8, Mrs Tan 40)
Workbook Think harder 4: Ben and Siti had $140 altogether. Ben spent 1/2 of his money and Siti spent $20. Then they had the same amount left. How much did Ben have at first?	$80 (Ben 2 units, Siti 1 unit + $20; 3 units = $120, 1 unit = $40)
Workbook Go further 5: Mei had $200 and Siti had $80. Mei spent 3 times as much money as Siti. Then they had the same amount left. How much money did Siti spend?	$60 (Siti spent 1 unit, Mei spent 3 units; the $120 difference is 2 units, so 1 unit = $60; both have $20 left)
Workbook Go further 6: Ravi had some stickers. He gave 1/2 of them and 4 more to Ali. Then he gave 1/2 of the stickers he had left and 4 more to Ben. He had 4 stickers left. How many stickers did he have at first?	40 (work backwards: (4 + 4) × 2 = 16, (16 + 4) × 2 = 40)

Workbook Go further 7: Ali and Ben had 140 stamps altogether. Ali gave away 1/3 of his stamps and Ben gave away 1/2 of his. Then they had the same number of stamps left. How many stamps did Ben give away?	40 (Ali 3 units, keeps 2 units; Ben keeps 2 units, so Ben had 4 units; 7 units = 140, 1 unit = 20)

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Ali has 50 marbles and Mei has 30. Ali gives Mei some marbles. Can the difference still be 20 afterwards?"*

"When does the total stay the same, and when does the difference stay the same?"

Discussion question	Listen for
Ali has 50 marbles and Mei has 30. Ali gives Mei some marbles. Can the difference still be 20 afterwards?	Yes. If Ali gives Mei 20 marbles, Ali has 30 and Mei has 50: the difference is 20 again, but now Mei has more. The total is 80 the whole time. Giving between two people keeps the total the same; both getting (or spending) the same amount keeps the difference the same.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Ali had 35 stickers and Siti had 25. How many stickers must Ali give Siti so that Ali has 1/2 as many stickers as Siti?	15 (the total stays 60; 3 units = 60, so Ali has 1 unit = 20 at the end: 35 − 20 = 15)

SAY *"Today you drew models with a before and an after, and you found the part that stays the same. Next time a story changes, ask yourself: what did NOT change?"*

"Show me your exit ticket and collect your stamp."

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "You drew the before and after bars side by side. That is how you saw what stayed the same."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

Make up your own before-and-after story where the total stays the same and the answer is 24. Swap with a partner and solve each other's story with a model.

Answer key (workbook)

Q1 (a) $20, $10, $30 (book 1/3 of $60 = $20, remainder $40, pen 1/4 of $40 = $10, left $30) (b) Too little ($30 left is less than $36): Ravi had $72 (3 units = $36 left out of 6 units, 1 unit = $12) (c) $12 (book $24, pen $12) (d) 1/6 (1/4 of the 2/3 that was left)

Q2 (a) 84 (giving does not change the total) (b) Siti 48, Mei 36 (after: 7 units = 84, 1 unit = 12; Siti 36 + 12 = 48)

Q3 (a) No. If Ali gives 20, Ali has 30 and Ben has 50. Giving 10 makes them the same (40 each): Ali loses 10 and Ben gains 10, so the gap of 20 closes by 20. (b) 20 (total 80 = 8 units, 1 unit = 10; Ali 30, Ben 50)

Q4 (a) The difference: $24 (b) $22 (the difference $24 is 1 unit; Ali 2 units = $48, so $48 – $26 = $22)

Q5 (a) Blue (only red beads were added) (b) blue 4 units; 4 units; the whole box is 10 units (blue is 2 out of every 5, so 4 units = 2 parts and 1 part = 2 units) (c) 70 (red 3 units at first and 6 units now, so 3 units = 30 and 1 unit = 10) (d) 100

Story try-it (optional, Alcuin of York) 3 units + 1/2 unit = 3 1/2 units = 100 − 2 = 98. 1 unit = 98 ÷ 3 1/2 = 28 storks. Check: 3 × 28 = 84, half of 28 = 14, and 84 + 14 + 2 = 100.

Lesson 4: Triangle Areas

Week 2 (second lesson) • Area of Triangle • Workbook: Lesson 4

Objective	Find the area of a triangle as 1/2 × base × height, choosing a base and the height that meets it at a right angle (also when the height is outside the triangle); use “same base, same height” to explain equal areas; find areas of composite figures and shaded parts by splitting into triangles or by taking away.
Thinking skills	Spotting the height, ignoring lengths that are not needed, reasoning with same base and same height, splitting a figure, taking away from a rectangle, explaining why an area does not change
Materials	Square grid paper (cut-out page at the back of the workbook), scissors, ruler, set square or the corner of a card, mini whiteboard and marker, workbook, pencil.
Online set-up	The tutor shares the slides and clicks to draw heights and reveal each step. For the explore, children draw and cut their rectangle with the camera up as usual, and hold the cut pieces up to the camera to show they fit (or use the interactive version on the slide). Whiteboards are held up together for the quick questions.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *“Welcome back, inquirers! A triangle is half of something. Today we find out half of WHAT, and why a squashed, leaning triangle can have exactly the same area as a neat one.”*

“Notice it, test it, explain why. Today “test it” means cutting paper.”

DO Greet each child by name. Show the lesson title slide. Read “Today we will…” aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): Draw a rectangle 6 squares wide and 4 squares tall on squared paper. Count all the squares in it. Draw a line from one corner to the opposite corner and cut along it. Do the two pieces match exactly? What fraction of the rectangle is one piece?	24 squares. The two triangles match exactly (turn one round and it fits on the other), so each is 1/2 of the rectangle: 12 squares.
Puzzle 2 (thinking): ABCD is a rectangle 12 cm long and 5 cm tall. P is a point on the top side AB. Join P to C and to D. Does the area of triangle PCD change if P slides along AB? What is the area?	No, 30 cm². The base DC = 12 cm and the height from P down to DC is always 5 cm, so the area is 1/2 × 12 × 5 = 30 cm² wherever P is. A guess of "it changes" is fine: we test it in the explore.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked." Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Half or not? Three rectangles each have one shaded triangle: one cut along a diagonal, one with its top corner on the top side, one from a corner to the middle of a side. Vote: is the shaded part exactly half?

SAY *"Here are three rectangles, each with a shaded triangle. Thumbs up if you think the triangle is exactly half of the rectangle, thumbs down if not. Decide for each one before I click."*

"For the one that is not half: how much of the rectangle is it? How do you know?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Area of a rectangle 7 cm by 6 cm?	42 cm²
Half of 42 cm²?	21 cm²
1/2 × 9 × 4 = ?	18
A square has an area of 49 cm². How long is each side?	7 cm
1/2 × 15 × 8 = ?	60

9–21 min Explore (in pairs, with objects)

SAY *"You will draw a rectangle 8 cm by 6 cm and a triangle inside it with its top corner on the top side. Predict: what fraction of the rectangle is the triangle? Does it matter where the top corner is?"*

DO 1. Draw a rectangle 8 cm by 6 cm on grid paper.

2. Mark any point on the top side. Join it to the two bottom corners.

3. Cut out the triangle and the two leftover pieces.

4. Do the two pieces cover the triangle exactly? Try again with a different point.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: fold each leftover piece along a grid line and lay it on the triangle.

Pieces do not fit: check that your top corner is exactly on the top side and the bottom corners are the rectangle's corners.

Finished early: move the top corner right to the end of the top side. What kind of triangle is it now? Is it still half?

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): On squared paper draw THREE different triangles that each have an area of exactly 12 square units. Write the base and the height of each one. What do all three have in common? (Answer: For example base 6 and height 4, base 8 and height 3, base 12 and height 2, base 4 and height 6. Each time base × height = 24, which is double the area (12 × 2). Any triangle on the same base and height, leaning or not, is also fine.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Do the two leftover pieces cover the triangle exactly?	Yes, so the triangle is half of the rectangle.
What is the area of the triangle?	24 cm² (half of 8 × 6 = 48)
Does it matter where the point on the top side is?	No: the base is always 8 cm and the height is always 6 cm.

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"This triangle has sides 5, 12 and 13. Which two do I multiply? The height must meet the base at a right angle. I look for the square corner: 12 and 5 meet there. 13 is slanting, so I do not use it. 1/2 × 12 × 5 = 30."*

"Here the top corner is past the end of the base. I extend the base with a dashed line and drop the height to it: 12 cm. The base is still only 11 cm: the dashed part is not part of the triangle. 1/2 × 11 × 12 = 66."

"All three triangles sit on the bottom of the same rectangle with their top corner on the top side. Same base, 14 cm. Same height, 9 cm. So they all have the same area, 63 cm², even though they look different."

Worked example on the slides	Answer
Height outside the triangle. A triangle has a base of 11 cm. Its top corner is past the end of the base. The height to the base extended is 12 cm, and its sides are 13 cm and 20 cm. What is its area?	66 cm² (1/2 × 11 × 12; the 13 cm and 20 cm sides are not needed).
Same base, same height. A rectangle is 14 cm by 9 cm. Three triangles each use the bottom side as the base, with the top corner at different points on the top side. What is the area of each triangle?	63 cm² each (1/2 × 14 × 9), half of the rectangle.

SAY *"The key idea today is: 1/2 × base × height. Area of a triangle = 1/2 × base × height. The height meets the base at a right angle, even when it is outside the triangle. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Using a slanting side as the height (1/2 × 14 × 13 for a 13-14-15 triangle).
- Forgetting the 1/2 and giving the area of the whole rectangle.
- For a height outside the triangle, measuring the base all the way to the foot of the height (adding the dashed extension to the base).

33–37 min Brain break

DO Triangle stretch. Make a triangle with your arms above your head. Now make a right angle with one arm. Lean your triangle to the left and to the right: did its height change?

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Liu Hui, 3rd century CE, China (kingdom of Wei)." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today you find triangle areas, and Liu Hui shows that it helps to ask why a rule works, not just how to use it. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): Rectangle area = 6 × 8 = 48 cm². One triangle is half of that: 24 cm². SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say "Pairs first. Talk before you write."

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

"After question 3 you may choose: carry on, or try Think harder. You can swap at any time."

DO Circulate and ask:

- Ask: "Point to the height. How do you know it meets the base at a right angle?"
- Ask: "Which length did you not need? Why not?"
- Ask: "Did you split the figure or take away? Could you do it the other way to check?"

IF STUCK Hint 1: "Find the square corner (the right angle). The two lines that make it are a base and a height."

Hint 2: "If there is no square corner, look for a dashed line drawn at a right angle to a side."

Hint 3: "Box the shape in a rectangle. Find the rectangle's area, then take away the white triangles."

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Triangle Areas (Find all the ways)	First step for everyone: Easy first step: draw a right-angled triangle on squared paper with short sides 3 cm and 4 cm. What is its area? Then look for others with area 6 cm². (a) On square grid paper (1 square = 1 cm), draw right-angled triangles whose two short sides are along the grid lines and are whole numbers of cm, and whose area is exactly 6 cm². Find ALL of them (a triangle turned round is the same triangle). How do you know you have them all? (b) Now keep the base 4 cm along the bottom. Mark where the top corner can go so that the area is still 6 cm². What do you notice about all the possible top corners? (c) What if the area is 10 cm²? 12 cm²? How many right-angled triangles can you find each time? Which area gave the most?
What to look for / model answer	Part 1: three triangles, with short sides 1 and 12, 2 and 6, 3 and 4 (the two short sides multiply to 12, which is double the area; checked by listing every pair). Part 2: base 4 means height 3 cm, so the top corner can be anywhere on a straight line 3 cm above and parallel to the base (same base, same height, same area), not just above the end of the base. Part 3: 10 cm² gives 3 (1 and 20, 2 and 10, 4 and 5); 12 cm² gives 4 (1 and 24, 2 and 12, 3 and 8, 4 and 6); 12 gave the most because 24 has the most factor pairs. Look for: a systematic list, the doubling idea, and the parallel-line observation.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: The area of a right-angled triangle is 24 cm². Its two short sides (the ones that make the right angle) differ by 2 cm. How long are the two short sides?	6 cm and 8 cm. Base × height = 48, so find two numbers that multiply to 48 and differ by 2: 6 × 8 = 48. Check: 1/2 × 6 × 8 = 24.
Workbook Think harder 2: A right-angled triangle has sides of 12 cm, 16 cm and 20 cm. What is its area?	96 cm² (1/2 × 12 × 16)
Workbook Think harder 3: A triangle and a square have the same area. The square has sides of 6 cm. The triangle has a base of 9 cm. What is the height of the triangle?	8 cm (the square is 36 cm², and 1/2 × 9 × 8 = 36).
Workbook Think harder 4: A triangle and a rectangle have the same base, 12 cm, and the same height, 7 cm. What is the difference between their areas?	42 cm² (84 – 42)
Workbook Go further 5: A right-angled triangle has sides of 15 cm, 20 cm and 25 cm. What is its height when the 25 cm side is the base?	12 cm (area 1/2 × 15 × 20 = 150 cm², and 1/2 × 25 × 12 = 150)
Workbook Go further 6: Two squares stand side by side on a line, touching: a big square of side 10 cm and a small square of side 6 cm. Join the bottom-left corner of the big square, the top-right corner of the big square and the top-right corner of the small square to make a triangle. What is its area? Try again with a small square of side 4 cm. What do you notice?	50 cm² both times: it is always half of the big square (box it in a rectangle and take away three right-angled triangles: 160 – 50 – 12 – 48 = 50, and 140 – 50 – 12 – 28 = 50)
Workbook Go further 7: Triangle ABC has an area of 54 cm². D is a point on BC with BD = 4 cm and DC = 8 cm. What is the area of triangle ABD?	18 cm² (the height is 9 cm because 1/2 × 12 × 9 = 54; ABD has base 4 cm and the same height)

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Ben says a triangle with longer sides always has a bigger area. Is he right?"*

"How can two triangles that look very different have the same area?"

Discussion question	Listen for
Ben says, "A triangle with longer sides always has a bigger area." Is he right?	No. Two triangles with the same base and the same height have the same area, even if one leans over and has much longer sides. The area depends on the base and the height, not on the slanting sides. A very long, thin triangle can even have a smaller area than a short, wide one.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A triangle has an area of 42 cm². Its base is 12 cm. What is its height?	7 cm (1/2 × 12 × height = 42, so 6 × height = 42)

SAY *"Today you found the height (sometimes outside the triangle), and you saw why triangles on the same base with the same height have the same area. That is explaining why."*

"Show me your exit ticket and collect your stamp."

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "You split the shape into triangles when it was too big. That is a strong strategy."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

On a grid, draw as many different triangles as you can with area 6 cm² using a base of 4 cm. Where can the top corner go? (Anywhere on the line 3 cm above the base.)

Answer key (workbook)

Q1 (a) 24 cm² (1/2 × 8 × 6) (b) 84 cm² (1/2 × 14 × 12) (c) 36 cm² (1/2 × 9 × 8) (d) 13 cm and 15 cm: they are slanting sides, not at a right angle to the base.

Q2 (a) 30 cm² (b) 30 cm² (c) Both have base DC = 10 cm and height 6 cm (P and Q are on AB, 6 cm above DC).

Q3 (a) 5 cm; 35 cm² (1/2 × 14 × 5) (b) 3 cm; 21 cm² (1/2 × 14 × 3) (c) 56 cm²

Q4 (a) 96 cm² (b) ADE 16 cm², EBF 16 cm², FCD 24 cm² (c) 40 cm² (96 – 56)

Q5 (a) 24 cm² (b) 4 cm (c) any base and height that multiply to 48, e.g. base 16 cm and height 3 cm, or base 6 cm and height 8 cm (or base 8 cm and height 6 cm leaning over)

Story try-it (optional, Liu Hui) Rectangle area = 6 × 8 = 48 cm². One triangle is half of that: 24 cm².

Lesson 5: Volume Vault

Week 3 (first lesson) • Volume • Workbook: Lesson 5

Objective	Find the volume of a cuboid by counting layers of 1-cm cubes and by length × breadth × height; list every cuboid that can be built from a number of cubes; solve water-tank problems (cm³ and litres, new water levels, the rise when an object sinks, pouring from one tank into another); and count the painted faces of a cube cut into small cubes.
Thinking skills	Make a list in order, draw a diagram to scale, work backwards (height = volume ÷ base area), look for a pattern (painted cubes), check with a second method
Materials	About 30 one-centimetre cubes (or dice, Lego bricks, sugar cubes), a ruler, squared paper, mini whiteboards. Tutor: a clear plastic box with a ruler taped to its side, water and a stone for the demonstration.
Online set-up	The tutor shows the water rise live on camera: a clear box, a ruler taped to the side, a stone dropped in. Children use dice or Lego for the Explore; with no cubes they draw layers on squared paper. Whiteboards are held up to the camera.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *"Welcome to the Volume Vault, inquirers! Today everything is made of tiny 1-centimetre cubes: boxes, tanks, even water. If we can count cubes cleverly, we can work out how much a tank holds, how high the water goes and what happens when a stone sinks in."*

"Watch this clear box. I am going to drop a stone in. Before I do: what will happen to the water level, and could we work out the stone's volume from it?"

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): A box is 4 cubes long, 3 cubes wide and 2 cubes tall. Build it with cubes, or draw one layer and then the next layer. How many cubes fill the box?	24 cubes: one layer is 4 × 3 = 12 cubes and there are 2 layers.
Puzzle 2 (thinking): A box has a base 5 cubes long and 4 cubes wide. Draw the base and count the cubes in ONE layer. How many layers are needed to hold 60 cubes altogether?	3 layers. One layer is 5 × 4 = 20 cubes, and 60 ÷ 20 = 3. (Counting layers of 20 on a drawing also works.)

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked."

Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Missing Edge Match. Six cuboid cards each show the volume and two edges. Teams find the third edge and write the division they used (60 ÷ (5 × 4) = 3). Show answers together on three, two, one.

SAY *"Missing Edge Match! Each card shows a cuboid's volume and two of its edges. Find the third edge and write the division you did. We show our answers together on three, two, one. Nobody shouts, and there is no prize for being first."*

"First card: 60 cm³, 5 cm and 4 cm. What is the third edge? How do you know?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
A cuboid is 7 cm by 5 cm by 2 cm. What is its volume?	70 cm³
1 ℓ = ? cm³	1000 cm³
3.2 ℓ = ? ml	3200 ml
A cube has a volume of 125 cm³. How long is each edge?	5 cm
4500 cm³ of water = ? ℓ	4.5 ℓ

9–21 min Explore (in pairs, with objects)

SAY *"You have 24 one-centimetre cubes. You must use all of them to build one cuboid. How many different cuboids do you think you can build? Write a guess: 2? 5? 10?"*

DO 1. Take 24 one-centimetre cubes (or draw them on squared paper).

2. Build a cuboid that uses all 24 cubes. Write its length, breadth and height.

3. Build as many different cuboids as you can. Turning one round does not make a new one.

4. How do you know you have found them all? Which one is most like a cube?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with a flat cuboid, 1 cube high. Then try 2 cubes high: how many cubes are in each layer then?

Not sure if two are different: "Turn it round. If it is the same three edges, it is the same cuboid."

Finished early: which of your cuboids is most like a cube? Can any of them be a cube? Why not?

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): Try other numbers of cubes: 12, 16, 18, 20, 24 and 30. Which number gives the MOST different cuboids? Which numbers up to 30 give only ONE? (Answer: 24 gives the most, 6 cuboids (12, 16, 18 and 20 give 4 each; 30 gives 5). Only the prime numbers 2, 3, 5, 7, 11, 13, 17, 19, 23 and 29 (and 1) give just one cuboid, a single row: 1 × 1 × n. Checked by listing every case.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many different cuboids can you build from 24 cubes?	6: 1 × 1 × 24, 1 × 2 × 12, 1 × 3 × 8, 1 × 4 × 6, 2 × 2 × 6, 2 × 3 × 4.
How do you know you have found them all?	List in order with the shortest edge first: shortest edge 1 (1 × 1, 1 × 2, 1 × 3, 1 × 4; after that the pairs repeat), then shortest edge 2 (2 × 2 × 6, 2 × 3 × 4). Shortest edge 3 would need 3 × 3 × something, and 24 ÷ 9 is not whole.
Do they all have the same volume?	Yes: 24 cm³ each. Same volume, different shapes.

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"A cuboid 5 cm by 3 cm by 4 cm. One layer on the bottom is 5 rows of 3: 15 cubes. There are 4 layers, so 15 × 4 = 60 cubes. That is 60 cm³. Length × breadth is the layer, and × height counts the layers."*

"The water in the tank is a cuboid too: 40 by 25 by 12. That is 12 000 cm³. 1000 cm³ is 1 litre, so 12 litres. Now I pour in 8 litres: that is 8000 cm³. The base is 40 × 25 = 1000 cm², so each centimetre of height holds 1000 cm³. 8000 ÷ 1000 = 8 cm higher."

"A stone goes in and the water goes up 3 cm. The water did not change: the stone is pushing it up. The new slice of height is 30 by 20 by 3 = 1800 cm³. That slice is exactly the stone's volume."

Worked example on the slides	Answer
Water in a tank: 40 cm by 25 cm by 30 cm, water 12 cm deep. The water is a cuboid: 40 × 25 × 12 = 12 000 cm³ = 12 ℓ. 8 ℓ more = 8000 cm³. Base area 40 × 25 = 1000 cm², so the water rises 8000 ÷ 1000 = 8 cm. New depth 12 + 8 = 20 cm. To fill it: 40 × 25 × 10 = 10 000 cm³ = 10 ℓ more.	12 ℓ of water; after 8 ℓ more the water is 20 cm deep; 10 ℓ more fills the tank.
A stone sinks: tank base 30 cm by 20 cm, water rises from 10 cm to 13 cm. The water does not change. The stone pushes it up. The new slice is 30 × 20 × 3 = 1800 cm³: that is the stone's volume. Backwards: a 600 cm³ stone would make it rise 600 ÷ 600 = 1 cm.	The stone is 1800 cm³. Rise = object volume ÷ base area.

SAY *"The key idea today is: volume = base area × height. A cuboid, or the water in a tank, holds base area × height cubic centimetres. Anything added or sunk in raises the water by its volume ÷ base area. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Mixing units: dividing litres by a base area in cm² without changing litres to cm³ first (8 ℓ = 8000 cm³).
- Using the tank's full height instead of the water height, or giving the new water level when the question asks for the rise (or the other way round).
- Painted cubes: counting the corner cubes again on the edges, or using n instead of n − 2 for the cubes along each edge.

33–37 min Brain break

DO More or less than a litre? I name an object: a can of drink, a bathtub, a teacup, a fish tank, a big water bottle. Stand tall if it holds more than 1 ℓ, crouch if less.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Archimedes, about 287 to about 212 BC, Syracuse, Sicily (ancient Greek world)." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today you work out volumes of cuboids and water levels, and the bath story is about using the water a solid pushes aside to measure volume. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): The water rises as a cuboid 20 cm × 10 cm × 2 cm = 400 cm³. So the stone has volume 400 cm³ (400 ml). Keep the label: A story often told about Archimedes (the bath and the wreath) is from a writer about two hundred years later; historians doubt the details. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say “Pairs first. Talk before you write.”

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

“After question 3 you may choose: carry on, or try Think harder. You can swap at any time.”

DO Circulate and ask:

- Ask: “Which number is the water height, and which is the tank height?”
- Ask: “Is that in litres or in cm³? Change it before you divide.”
- Ask: “How do you know your list is complete? What did you keep fixed?”

IF STUCK Hint 1: “Draw the tank. Shade the water. Write every length you know on the picture.”

Hint 2: “What is the base area? Each 1 cm of height holds that many cm³.”

Hint 3: “Is the object completely under the water? If not, the water only has the space around it.”

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Volume Vault (Find all the ways)	First step for everyone: Easy first step: list every box that holds exactly 12 cubes (for example 1 × 1 × 12). Then do part (a). (a) A shop wants a box that holds exactly 48 small cubes of sugar, packed with no gaps. The box is a cuboid with whole-number edges in cubes. Find ALL the different boxes (a 2 by 4 by 6 box is the same as a 6 by 4 by 2 box). Make a list in order. (b) Which box would you choose to sell, and why? Which is most like a cube? (c) What if the shop packs 64 cubes? How many different boxes now? Is there a number of cubes that gives only ONE box shape?

What to look for / model answer	Part 1: nine boxes: 1×1×48, 1×2×24, 1×3×16, 1×4×12, 1×6×8, 2×2×12, 2×3×8, 2×4×6, 3×4×4 (brute-forced over every possible edge). Part 2: open; 3×4×4 is closest to a cube. Part 3: 64 cubes gives seven boxes (1×1×64, 1×2×32, 1×4×16, 1×8×8, 2×2×16, 2×4×8, 4×4×4); 64 has a cube-shaped box (4×4×4) but several others too. A prime number of cubes (2, 3, 5, 7, 11 ...) gives exactly ONE box, 1×1×n, and so does 1 cube. Look for: ordered list by smallest edge, factor pairs, and the reason a prime gives one box.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: How many cubes with 2-cm edges can be packed in neat layers into a box 9 cm long, 6 cm wide and 5 cm high?	24 (4 along, 3 across, 2 layers; not 270 ÷ 8, because the cubes cannot be cut)
Workbook Think harder 2: A tank is 1/3 full of water. After 12 ℓ more is poured in, it is 3/5 full. How many litres does the tank hold when full?	45 ℓ (3/5 − 1/3 = 4/15 of the tank is 12 ℓ, so 1/15 is 3 ℓ)
Workbook Think harder 3: A cuboid has a square base and a height of 5 cm. Its volume is 245 cm³. How long is each side of the base?	7 cm (base area 245 ÷ 5 = 49 cm², and 7 × 7 = 49)
Workbook Think harder 4: A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have no paint at all?	6 (the inside block is 1 × 2 × 3)
Workbook Go further 5: Three faces of a cuboid have areas 12 cm², 15 cm² and 20 cm². What is the volume of the cuboid?	60 cm³ (edges 3 cm, 4 cm and 5 cm: 3 × 4 = 12, 3 × 5 = 15, 4 × 5 = 20)
Workbook Go further 6: A cuboid 3 cm by 4 cm by 5 cm is built from 1-cm cubes and painted on the outside. How many cubes have exactly 2 painted faces? How many have exactly 1?	24 with 2 faces (4 × 1 + 4 × 2 + 4 × 3 along the edges); 22 with 1 face (60 − 8 corners − 24 − 6 inside)
Workbook Go further 7: A tank has a base 30 cm by 20 cm and water 10 cm deep. A block 10 cm by 10 cm by 15 cm is stood upright on the bottom. How deep is the water now?	12 cm (the water now covers 600 − 100 = 500 cm² of floor: 6000 ÷ 500 = 12 cm; the block is 15 cm tall, so it sticks out. Not 12.5 cm!)

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Ali says a 1000 cm³ stone always makes the water rise 1 cm. Is he right?"*

"Why does a big tank's water level rise only a little when a stone goes in?"

Discussion question	Listen for
Ali says: "A 1000 cm³ stone always makes the water rise 1 cm." Is he right? Explain.	Ali is right only when the base area is 1000 cm² (for example 40 cm by 25 cm). The rise is the stone's volume ÷ base area: in a 20 cm by 10 cm tank (200 cm²) the same stone makes the water rise 1000 ÷ 200 = 5 cm. A wide tank spreads the extra volume over a big base, so the rise is small. (And if the stone sticks out of the water, only the part under water counts.)

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A tank has a base 30 cm by 25 cm and water 8 cm deep. A rock is put in and sinks. The water rises to 10 cm. (a) What is the volume of the rock? (b) How many litres of water are in the tank?	(a) 1500 cm³ (30 × 25 × 2) (b) 6 ℓ (30 × 25 × 8 = 6000 cm³)

SAY *"Today you counted cubes in layers, listed every cuboid in order, and worked backwards from a water level to a volume. Notice it, test it, explain why: you did all three."*

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "Thinking of the water as layers of cubes got you unstuck."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

Find every cuboid with whole-number edges and a volume of 60 cm³. (There are 10, from 1 × 1 × 60 to 3 × 4 × 5.) Which one is most like a cube?

Answer key (workbook)

Q1 (a) 24 cm³ (4 × 3 × 2) (b) 40 cubes (the smallest cube it fits in is 4 × 4 × 4 = 64 cubes; 64 − 24 = 40)

Q2 (a) 8 cuboids: 1 × 1 × 36, 1 × 2 × 18, 1 × 3 × 12, 1 × 4 × 9, 1 × 6 × 6, 2 × 2 × 9, 2 × 3 × 6, 3 × 3 × 4 (b) only 3 × 3 × 4 (every other one has an edge longer than 5 cm)

Q3 (a) 27 ℓ (50 × 30 × 18 = 27 000 cm³) (b) 33 ℓ (50 × 30 × 22 = 33 000 cm³) (c) 12 cm (9000 ÷ 1500 = 6 cm lower; 18 − 6 = 12)

Q4 (a) 5000 cm³ (5 × 10 × 10 × 10) (b) 5 cm (5000 ÷ 1000) (c) 20 cm (d) 3000 cm³ (40 × 25 × 3)

Q5 3 faces: 8 (corners) 2 faces: 24 (12 edges × 2) 1 face: 24 (6 faces × 4) no paint: 8 (2 × 2 × 2 inside). Check: 8 + 24 + 24 + 8 = 64

Q6 (a) 14.4 ℓ (30 × 20 × 24 = 14 400 cm³) (b) 12 cm (14 400 ÷ 1200) (c) 8 cm (14 400 ÷ (600 + 1200) = 8)

Story try-it (optional, Archimedes) The water rises as a cuboid 20 cm × 10 cm × 2 cm = 400 cm³. So the stone has volume 400 cm³ (400 ml).

Lesson 6: Rate Racers

Week 3 (second lesson) • Rate • Workbook: Lesson 6

Objective	Find a rate for 1 unit and use it to scale up or work backwards in a rate table; compare buys by the price of one item; add rates when taps or printers work together and take away a leak; work out charges “for every half hour or part of it”; and compare two price plans with a table.
Thinking skills	Find the value of 1 unit, make a table, compare, guess and check with a table, draw a timeline, check with a second method
Materials	Mini whiteboards, a calculator for checking only, a stopwatch (or phone), a few real price tags or supermarket flyers if you have them.
Online set-up	Children hold whiteboards up to the camera; the tutor fills the rate tables on the slides. Pairs work in breakout rooms for the Explore. The brain break uses a phone stopwatch.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *“Welcome, Rate Racers! Today is about “how much for one”: one minute, one kilogram, one can. Once you know the rate for one, you can predict anything: how many pages in an hour, which pack is cheaper, when the tank will be full.”*

“Shops, taps, printers and car parks all run on rates. By the end you will spot the trap in “Buy 10, get 2 free!”

DO Greet each child by name. Show the lesson title slide. Read “Today we will…” aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): A tap puts 12 litres into a tank in 4 minutes. Draw 4 buckets and share the 12 litres. How many litres go in each minute? How many litres go in during 6 minutes?	3 litres each minute, so 18 litres in 6 minutes (6 × 3).
Puzzle 2 (thinking): Ali reads 12 pages in 4 minutes. Mei reads 20 pages in 5 minutes. Who reads faster? How can you be sure?	Mei. Ali reads 3 pages a minute (12 ÷ 4) and Mei reads 4 pages a minute (20 ÷ 5). Comparing the amount for 1 minute makes it fair.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked." Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Better Buy Showdown. First pair: pens, 4 for $2.00 or 10 for $4.00 (50¢ and 40¢ each). Point to the better buy, then prove it with the price of 1. Tutor extension if the group is ready: juice (1.5 ℓ for $3.00 or 2 ℓ for $3.80) and eggs (10 for $3.20 or 12 for $3.96).

SAY *"Better Buy Showdown! Two packs of the same thing. Point left or right: which is the better buy? Then prove it."*

"Pens: 4 for $2.00, or 10 for $4.00. Point now. Who can prove it? Volunteers only, please. The price of ONE pen each time is the fair way to compare."

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
4 kg of rice cost $10. How much is 1 kg?	$2.50
A printer prints 30 pages a minute. How many pages in 6 minutes?	180 pages
A tap lets out 450 ml in 3 minutes. How many ml a minute?	150 ml
Cheese costs $1.20 for 100 g. How much is 1 kg?	$12
Ali types 40 words a minute. How long for 200 words?	5 minutes

9–21 min Explore (in pairs, with objects)

SAY *"Printer A prints 72 pages in 4 minutes. Printer B prints 50 pages in 2 minutes. Which is faster? Now both print a 430-page job together. Guess the time before you work it out."*

DO 1. Printer A prints 72 pages in 4 minutes. Printer B prints 50 pages in 2 minutes.

2. Make a rate table for each printer: 1, 2, 4 and 10 minutes.

3. Which printer is faster? Show how you know.

4. Predict, then work out: how long do both printers together take to print 430 pages?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: how many pages does each printer print in 1 minute? Start the table with 1 minute.

Not sure how to compare: "Compare the same number of minutes for both printers."

Finished early: how many pages should go to each printer so both finish at the same time?

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): Printer A prints 18 pages a minute and printer B prints 25 pages a minute. Together they print 430 pages in 10 minutes. How many pages should each printer be given so that both finish at exactly the same time? (Answer: Printer A gets 180 pages (18 × 10) and printer B gets 250 pages (25 × 10); 180 + 250 = 430. If both work for the whole 10 minutes they finish together.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Which printer is faster? How do you know?	B: 25 pages a minute; A: 18 pages a minute. (Or compare 4 minutes: A 72, B 100.)
How long do both together take for 430 pages?	10 minutes: together 18 + 25 = 43 pages a minute, and 430 ÷ 43 = 10.
Why is the rate for 1 minute so useful?	We can compare the printers fairly, multiply to scale up and divide to find a time.

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"5 kg of rice cost $9.50. I want the price of 1 kg first: $9.50 ÷ 5 = $1.90. Now anything is easy: 3 kg is 3 × $1.90 = $5.70. And backwards: $13.30 is 1330 cents and 1 kg costs 190 cents, so 1330 ÷ 190 = 133 ÷ 19 = 7 kg."*

"Pens: 3 for $1.95 is 65¢ each. 5 for $3.10 is 62¢ each. So the 5-pack is the better buy, if I need that many. I check another way: 15 pens cost 5 × $1.95 = $9.75 or 3 × $3.10 = $9.30."

"Two taps fill 9 ℓ and 6 ℓ each minute. Together, every minute, 15 ℓ go in. 180 ÷ 15 = 12 minutes. A leak takes out 3 ℓ a minute, so only 12 ℓ stay in each minute: 180 ÷ 12 = 15 minutes."

Worked example on the slides	Answer

Better buy: pens at 3 for $1.95 or 5 for $3.10. Find the price of 1 pen at each shop. Check another way: 15 pens cost $9.75 or $9.30.	The 5-pack is the better buy (62¢ a pen). But if Mei needs only 3 pens, the 3-pack costs less.
Taps together: a 180 ℓ tank, taps of 9 ℓ and 6 ℓ a minute. Together: 9 + 6 = 15 ℓ a minute, so 180 ÷ 15 = 12 minutes. A leak lets out 3 ℓ a minute: 15 − 3 = 12 ℓ stay in each minute. 180 ÷ 12 = 15 minutes.	12 minutes; with the leak, 15 minutes. Add the rates, never the times.

SAY *"The key idea today is: find the rate for 1. Find how much for 1 unit (1 minute, 1 kg, 1 can). Multiply to scale up; divide to find how many or how long. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Adding or averaging the times when taps or printers work together (20 min and 30 min gives 50 min or 25 min) instead of adding the rates.
- Choosing the better buy by the pack price, or dividing a free-gift deal by the cans paid for instead of all the cans received.
- Charges "for every half hour or part of it": dividing exactly instead of rounding up to the next half hour, or charging the first hour twice.

33–37 min Brain break

DO Rate race! How many star jumps can you do in 10 seconds? Predict your 1-minute total (× 6). Then test for 30 seconds: does your rate stay the same?

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Mary Jackson, 1921–2005, United States." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: A wind tunnel test is full of rates: how far air or a plane moves in each hour or each second, just like the rates you work with today. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): 840 ÷ 3 = 280 km in 1 hour (a rate of 280 km per hour). In 5 hours: 280 × 5 = 1400 km. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say "Pairs first. Talk before you write."

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

"After question 3 you may choose: carry on, or try Think harder. You can swap at any time."

DO Circulate and ask:

- Ask: "What is the rate for 1? One minute, one can, one kilogram?"
- Ask: "How many cans do you really get in that deal?"
- Ask: "Each minute, how much goes in and how much comes out?"

IF STUCK Hint 1: "Find the amount for 1 unit first."

Hint 2: "Make a table: 1 minute, 2 minutes, 5 minutes ... or draw a timeline."

Hint 3: "Working together? Add the rates, not the times."

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Rate Racers (Find all the ways)	First step for everyone: Easy first step: tap A runs for 6 minutes. How many litres go in? How many litres are still needed, and how many minutes of tap B give that? Then do part (a). (a) Tap A puts 2 litres a minute into an empty tank. Tap B puts 3 litres a minute. The taps are used ONE AT A TIME, each for a whole number of minutes (0 is allowed). Find ALL the ways to put exactly 24 litres into the tank. Make a table. (b) Which way takes the shortest total time? Why does the faster tap win? (c) What if the tank needs 25 litres? How many ways now? What changes and what stays the same?
What to look for / model answer	Part 1: five ways (minutes on tap A, minutes on tap B): (12, 0), (9, 2), (6, 4), (3, 6), (0, 8). Each time B is on 2 minutes more, A must be on 3 minutes fewer (6 litres swapped). Part 2: B only, 8 minutes (total times are 12, 11, 10, 9, 8): B puts in 3 litres a minute, more than A, so less time. Part 3: 25 litres gives four ways: (11, 1), (8, 3), (5, 5), (2, 7); shortest is 9 minutes (2 and 7). The swap rule (3 minutes of A for 2 minutes of B, always 6 litres) stays the same. The number of ways changes because 25 is odd, so tap B must be on an odd number of minutes (1, 3, 5 or 7); this was checked by listing all pairs. Look for: an ordered table and noticing the pattern in the table.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: A tap fills a tank in 12 minutes at 15 ℓ a minute. When a second tap is also turned on, the tank fills in 9 minutes. How many litres a minute does the second tap fill?	5 ℓ a minute (the tank holds 180 ℓ; 180 ÷ 9 = 20 ℓ a minute together; 20 − 15 = 5)
Workbook Think harder 2: Eggs are sold in boxes of 10 for $3.30 and boxes of 6 for $2.10. Siti needs at least 28 eggs. What is the least she can pay?	$9.60 (one box of 10 and three boxes of 6 make 28 eggs; three boxes of 10 cost $9.90)
Workbook Think harder 3: A car park charges $1.50 for the first hour and $1 for every half hour or part of a half hour after that. How much does it cost to park from 10.15 a.m. to 2.50 p.m.?	$9.50 (4 h 35 min: first hour, then 3 h 35 min is 8 half hours or parts)
Workbook Think harder 4: A train travels 150 km in 2 hours 30 minutes and does not change its speed. How far does it travel in 1 hour 45 minutes?	105 km. The speed is 150 ÷ 2.5 = 60 km an hour. 1 hour 45 minutes is 1.75 hours, and 60 × 1.75 = 105 km (60 + 45).
Workbook Go further 5: 3 cats eat 3 fish in 3 minutes. At this rate, how many cats are needed to eat 100 fish in 100 minutes?	3 cats (each cat eats 1 fish every 3 minutes, so the 3 cats eat 1 fish a minute: 100 fish in 100 minutes)
Workbook Go further 6: Candle A is 20 cm tall and burns down 2 cm every hour. Candle B is 14 cm tall and burns down 1 cm every hour. They are lit at the same time. After how many hours are they the same height? How tall are they then?	After 6 hours, both 8 cm tall (the gap of 6 cm closes by 1 cm each hour)
Workbook Go further 7: A tap fills an empty bath in 12 minutes. With the plug out, a full bath empties in 20 minutes. Ben turns on the tap but forgets the plug. How long does the bath take to fill?	30 minutes (pretend the bath holds 60 ℓ: 5 ℓ a minute in, 3 ℓ a minute out, 2 ℓ a minute stay: 60 ÷ 2)

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Is the biggest pack always the best buy? Give an example with numbers."*

"Where did you see a bigger pack that was NOT the better buy today?"

Discussion question	Listen for

Is the biggest pack always the best buy? Give an example with numbers.	No. Compare the price of 1: for example 3 pens for $1.95 is 65¢ a pen but 5 pens for $3.10 is 62¢ a pen (here the bigger pack is better); but if the 10-pack costs $7.50 and the 4-pack costs $2.80, the 4-pack is 70¢ a pen and the 10-pack is 75¢ a pen, so the small pack is the better buy. The better buy also depends on how many you really need.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Tap A fills 15 ℓ a minute and tap B fills 10 ℓ a minute. How long do both taps together take to fill a 200 ℓ tank? How long if a leak also lets out 5 ℓ a minute?	8 minutes (200 ÷ 25); with the leak 10 minutes (200 ÷ 20)

SAY *"Today you found the rate for 1 and raced with it: scaling up, working backwards, adding taps together and catching the free-gift trap. Notice it, test it, explain why."*

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "Making a table of 1, 2, 3 minutes got you unstuck."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

Make a better-buy poster: three pack sizes of the same snack where the biggest pack is NOT the best buy. Swap with a partner: can they find the best buy and prove it?

Answer key (workbook)

Q1 (a) Bottles: 16, 48, 80; minutes for 160 bottles: 10; 12 minutes: 192 bottles (b) 9.25 a.m. (400 ÷ 16 = 25 minutes)

Q2 (a) A 70¢, B 73¢, C 65¢ (Shop C gives 12 cans for $7.80) (b) Shop C (c) Three ways make exactly 20 cans: 1 deal from C (12 cans) + 2 packs from B = $7.80 + $5.84 = $13.64 (the cheapest); 2 packs from A + 2 packs from B = $8.40 + $5.84 = $14.24; 5 packs from B = $14.60. Any correct way with its correct cost is fine

Q3 (a) 20 minutes (240 ÷ 12) (b) 8 minutes (240 ÷ 30) (c) 10 minutes (240 ÷ 24) (d) 20 minutes (120 ÷ 6)

Q4 (a) 18 minutes (900 ÷ 50) (b) 20 minutes (P prints 150 pages alone in 5 min; 750 ÷ 50 = 15 min more) (c) 540 pages (both work 18 min: P 30 × 18 = 540, Q 20 × 18 = 360)

Q5 (a) $4.80 (3 h 35 min: first hour $1.20, then 2 h 35 min is 5 half hours and 5 min, so 6 lots of 60¢ = $3.60) (b) 3 hours ($3.60 – $1.20 = $2.40 = 4 half hours after the first hour)

Q6 (a) Gym A $50, $70, $90; Gym B $35, $70, $105 (b) 10 visits (c) Gym A, by $12 ($86 and $98)

Story try-it (optional, Mary Jackson) 840 ÷ 3 = 280 km in 1 hour (a rate of 280 km per hour). In 5 hours: 280 × 5 = 1400 km.

Lesson 7: Percentage Puzzles

Week 4 (first lesson) • Percentage • Workbook: Lesson 7

Objective	Find any percentage of an amount by building it from 10%, 5% and 1%; work out a discount and then 9% GST on the sale price; find a percentage of a remainder; and spot the classic percentage traps. (A preview of P6, on the Think Harder page: find a percentage increase or decrease by comparing the change with the original amount.)
Thinking skills	Asking "percentage of what?", building from 10% and 1%, bar models, making a table, comparing deals, explaining a trap
Materials	Mini whiteboard and marker, workbook, pencil, a hundred-square page (or squared paper), a calculator for checking only at the end of each page.
Online set-up	Children keep the workbook and a whiteboard in front of the camera. The tutor screen-shares the slides and draws bar models with the annotation tool. Calculators stay closed until the check-together slides.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *"Welcome back, inquirers! Look around any shop in Singapore and you will see percentages everywhere: 20% off, 9% GST, prices up 10%. Shops are very good at percentages. Today we make sure we are better."*

"Notice it, test it, explain why. Today the big question is always: percentage of WHAT?"

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): A 10 by 10 square has 100 small squares. Colour in 35 of them. How many are NOT coloured? What percentage is not coloured?	65 squares, which is 65%. Percent means out of 100.
Puzzle 2 (thinking): A jacket costs $100. The shop takes 10% off, and then takes another 10% off the NEW price. Is that the same as 20% off? What is the final price?	No. $100 → $90 → $81. 20% off $100 would be $80. The second 10% is taken from $90, not from $100.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked." Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Hundred-square snap. Three 10 by 10 squares have part shaded (30, 45 and 64 small squares). Write the percentage shaded and the percentage not shaded before the reveal.

SAY *"Here are three hundred-squares. Each one has 100 small squares. Write how much is shaded as a percentage, and how much is NOT shaded. Whiteboards up when I say go."*

"Did anyone find the "not shaded" part without counting? How?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
50% of $90	$45
10% of 350	35
25% of 48	12
1% of $700	$7
3 out of 20 as a percentage	15%

9–21 min Explore (in pairs, with objects)

SAY *"A bicycle costs $200. The price goes up by 10%. Later the new price goes down by 10%. What is the price now? Write your prediction before you work anything out."*

"Thumbs up if you think it is back to $200. Thumbs down if you think it is less. Sideways if you think it is more."

DO 1. A bicycle costs $200. Predict: if the price goes up 10% and then down 10%, what is it?

2. Work it out: up 10% first. Then take 10% of the NEW price off.

3. Now try it the other way: down 10% first, then up 10%.

4. Compare both answers with your prediction. What do you notice?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: what is 10% of $200? Add it on. Now what is 10% of the NEW price?

Got $200: check which amount you took 10% of the second time. Was it $200 or $220?

Finished early: try up 50% then down 50%. Is the gap bigger or smaller than with 10%?

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): A $200 price goes up 50% and then down 50%. What is the price now? Then try up 20% and down 20%. What do you notice about the final price each time? (Answer: Up 50% then down 50%: $200 → $300 → $150. Up 20% then down 20%: $200 → $240 → $192. The final price is always LESS than $200, because the "down" percentage is taken from the bigger new price.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Is the price back to $200?	No: $200 → $220 → $198.
Does the order matter?	No: $200 → $180 → $198 too.
Why is it $2 less?	The 10% taken off is 10% of $220 = $22, which is more than the $20 that was added.

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"I want 45% of $60. 10% is one tenth, so I divide by 10: $6. Half of that is 5%: $3. 45% is four 10%s and one 5%: $24 + $3 = $27."*

"98% is nearly all of it. So I find what is missing: 2%. 1% of $60 is 60 cents, so 2% is $1.20, and 98% is $60 − $1.20 = $58.80."

"For the fan I take off the discount first: 15% of $120 is $18, so the sale price is $102. The GST is 9% of $102, not of $120. 10% of $102 is $10.20, 1% is $1.02, so 9% is $9.18. I pay $111.18."

Worked example on the slides	Answer
Build it from 10%, 5% and 1%. Find 45% and 98% of $60. 10% = $6, 5% = $3, so 45% = $24 + $3. 1% = $0.60, so 98% = $60 − $1.20.	45% of $60 = $27; 98% of $60 = $58.80.
Discount first, then GST. A fan costs $120 before GST. It is sold at 15% off. Then 9% GST is added. 15% of $120 = $18, so the sale price is $102. 9% of $102 = $10.20 − $1.02.	GST $9.18; you pay $111.18.

SAY *"The key idea today is: percentage of WHAT?. Build any percentage from 10%, 5% and 1%. Before each step, ask: percentage of what? Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Adding percentages of different wholes: 20% of the money and then 25% of the REMAINDER is not 45% of the money.
- Thinking that going up by a percentage and then down by the same percentage brings you back to the start.
- Working out a percentage increase or decrease using the new amount instead of the original amount, or adding GST to the price before the discount and then taking the discount off the wrong amount.

33–37 min Brain break

DO Percent stretch. The tutor calls out a percentage: 100% is standing tall with arms up, 50% is hands on your head, 25% is hands on your knees, 0% is sitting down. Then 75%? (hands on shoulders).

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Adam Ries (also spelled Adam Riese), 1492–1559, Germany." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today you work out discounts, GST and price changes with percentages, the kind of everyday money sums that Ries taught ordinary people to do. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): 20% of $200 = $40, so the sale price is $200 – $40 = $160. 9% of $160 = $14.40, so the final price is $160 + $14.40 = $174.40. Keep the label: Sources give his birth year as 1492 or 1493. The sources I opened do not name 'percentages' specifically; they list prices, interest, and profit and loss. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say "Pairs first. Talk before you write."

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

"After question 3 you may choose: carry on, or try Think harder. You can swap at any time."

DO Circulate and ask:

- Ask: “Percentage of what? Point to the whole.”
- Ask: “Can you build that percentage from 10%, 5% and 1%?”
- Ask: “After the first change, what is the new whole?”

IF STUCK Hint 1: “Find 10% first by dividing by 10.”

Hint 2: “Draw a bar for the whole and cut it into 10 equal parts.”

Hint 3: “Write what the whole is BEFORE you work out each percentage.”

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Percentage Puzzles (Is it always true? Test and explain)	First step for everyone: Easy first step: on a $100 bag take 20% off (that is $20) and write the new price. Then take 10% off that new price. Then do part (a) the other way round. (a) A $100 bag is on sale. Shop 1 takes 20% off and then a further 10% off the new price. Shop 2 takes 10% off first and then a further 20% off the new price. Work out both prices. Is one cheaper? (b) Test two more pairs of discounts of your own (for example 50% and 10%, or 30% and 20%). Does the ORDER of two discounts ever change the final price? Is it always true? (c) Is 20% off then 10% off the same as 30% off? Test and explain. (Hint: after 20% off, you pay 4/5. After 10% off, you pay 9/10.)
What to look for / model answer	Part 1: both $72 (100 → 80 → 72 and 100 → 90 → 72). Part 2: the order never matters; checked by computer for every pair of whole-number percentages from 1% to 99%. Reason: after 20% off you pay 4/5 of the price and after 10% off you pay 9/10, and 4/5 of 9/10 is the same as 9/10 of 4/5, so the order makes no difference. Part 3: not the same: 20% then 10% is $72, but 30% off is $70. The second discount is taken from a smaller price, so it takes off less than 10% of $100. Look for: at least two extra tests, the fraction reason (or "the same two parts of the price are kept"), and the "percentage of WHAT" idea.

Think harder and Go further questions

Question	Answer

Workbook Think harder 1: At a hawker centre, a plate of chicken rice went up from $4 to $5. A month later, it went back down from $5 to $4. (a) Find the percentage increase from $4 to $5. (b) Find the percentage decrease from $5 to $4. (c) Both changes are $1. Why are the two percentages different? Hint: percentage change = the change out of the amount you START with.	(a) 25% ($1 out of $4) (b) 20% ($1 out of $5) (c) Each change is compared with its own starting amount: $1 out of $4 is a bigger part than $1 out of $5. (Percentage change is a P6 skill: a preview.)
Workbook Think harder 2: A box of cereal costs $15 in two shops. Shop X: "Buy 3, get 1 free!" Shop Y: "30% off every box!" Mei buys 4 boxes. (a) How much does she pay at each shop? Which is the better buy, and by how much? (b) Shop X's deal is really how many percent off?	(a) Shop X $45 (she pays for 3 boxes); Shop Y $42 (30% of $60 = $18 off). Shop Y is the better buy, by $3. (b) 25% off (1 box free out of 4, $15 off $60)
Workbook Think harder 3: A $60 shirt gets 20% off and a $40 hat gets 10% off. Then 9% GST is added to the total of the two sale prices. How much does Ali pay?	$91.56. Shirt $60 − $12 = $48. Hat $40 − $4 = $36. Total $84. GST = 9% of $84 = $7.56, so Ali pays $84 + $7.56 = $91.56.
Workbook Think harder 4: Siti puts $2000 in a bank that pays 3% simple interest each year, always on the $2000 she started with. After how many years is the total interest first MORE than $200? How much interest is that?	4 years, $240. Each year the interest is 3% of $2000 = $60. After 3 years it is $180 (not enough) and after 4 years it is $240.
Workbook Go further 5: A price tag says $109, GST included. GST is 9%. Ali says the GST is 9% of $109, which is $9.81. Is he right? How much GST is in the price?	No. $109 is 109% of the price before GST, so 1% is $1 and the GST is $9. (9% of $109 is wrong because GST is 9% of the price BEFORE GST.)
Workbook Go further 6: An $80 lamp is on sale in two shops. Shop P takes 25% off. Shop Q takes 10% off, and then a further 15% off the new price. In which shop is the lamp cheaper, and by how much?	Shop P, by $1.20. P: $80 − $20 = $60. Q: $80 − $8 = $72, then 15% of $72 = $10.80 off, $61.20.
Workbook Go further 7: Mrs Tan spent 40% of her money on groceries and 25% of the remaining money on a gift. She had $54 left. How much money did she have at first?	$120. Work backwards: $54 is 75% of the remainder, so the remainder is $72. $72 is 60% of her money, so 10% is $12 and 100% is $120.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

> **DO** Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?"

hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Which is better for the shopper: add 9% GST first and then take 20% off, or take 20% off first and then add 9% GST?"*

"Mei says 20% off and then 9% GST is the same as 11% off. Is she right?"

Discussion question	Listen for
Which is better for the shopper: add 9% GST first and then take 20% off, or take 20% off first and then add 9% GST?	They give exactly the same price. On $100: GST first gives $109, then 20% off gives $87.20; 20% off first gives $80, then GST gives $87.20. Each change multiplies the price, and the order of multiplying does not matter. Mei is not right: 11% off $100 is $89, not $87.20, because the 9% GST is worked out on the smaller sale price.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A $30 toy is sold at 20% off. Then 9% GST is added to the sale price. How much do you pay?	$26.16 (20% of $30 = $6, sale price $24; 9% of $24 = $2.16; $24 + $2.16 = $26.16)

SAY *"Today you found that percentages can trick you when the whole changes. Before every percentage, ask: of what? Well done, inquirers!"*

"Show me your exit ticket and I will stamp your inquiry journal."

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "Using 10% as a stepping stone got you unstuck."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or

H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

A shop raises every price by 25%, then has a "20% off everything" sale. Is the sale price more than, less than or the same as the old price? Try it with $40, $100 and $64. Can you explain why?

Answer key (workbook)

Q1 10%: $24 5%: $12 1%: $2.40 15%: $36 35%: $84 99%: $237.60. (a) 35% = 3 × 10% + 5% = $72 + $12 = $84. (b) 99% = 100% − 1% = $240 − $2.40 = $237.60.

Q2 (a) $128 (20% of $160 = $32) (b) $11.52 (10% of $128 = $12.80, 1% = $1.28; $12.80 − $1.28) (c) $139.52 (d) Shop A, by 48¢ ($140 − $139.52)

Q3 (a) $100 (25% of $80 = $20) (b) $80 (20% of $100 = $20) (c) No. The 20% is taken from $100, not from $80. 20% of $100 is $20, the same $20 that was added, so the price is back to $80, not $84.

Q4 (a) $24 (20% of $120) (b) $24 (25% of the $96 left) (c) $72 (d) 60%. The 25% is of the $96 left, not of $120, so she spent $48 of $120 = 40% and 60% is left ($72 out of $120).

Q5 Girls 24, boys 16. Glasses: girls 6 (25% of 24), boys 8 (50% of 16), whole class 14. (a) 35% (14 out of 40)

Think Harder Q1 (a P6 preview: percentage change) (a) 25% ($1 out of $4) (b) 20% ($1 out of $5) (c) A percentage change compares the change with the starting amount. $1 out of $4 is a bigger part than $1 out of $5.

Think Harder Q2 (a) Shop X $45 (pays for 3 boxes); Shop Y $42 (30% of $60 = $18 off). Shop Y is the better buy, by $3. (b) 25% (1 box free out of 4, $15 off $60)

Story try-it (optional, Adam Ries) 20% of $200 = $40, so the sale price is $200 − $40 = $160. 9% of $160 = $14.40, so the final price is $160 + $14.40 = $174.40.

Lesson 8: Angle Chasers

Week 4 (second lesson) • Angles, Triangles and Quadrilaterals • Workbook: Lesson 8

Objective	Use the angle facts (straight line 180°, point 360°, vertically opposite, triangle 180°, isosceles and equilateral triangles, parallelogram, rhombus, trapezium) to chase unknown angles through a figure, step by step, giving a reason for every step; and list every case when only one angle of an isosceles triangle is given.
Thinking skills	Angle-chase chains with reasons, using a diagram carefully, finding all cases, checking by adding, explaining
Materials	Paper, ruler, pencil, scissors, protractor, red and blue pencils, the dot-grid cut-out, mini whiteboards, workbook.
Online set-up	Children tear or cut the corners of a paper triangle and quadrilateral on camera. The tutor screen-shares the slides and draws on the figures with the annotation tool. Children keep a red pencil for the angle they are chasing.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *"Welcome back, inquirers! Today we are detectives again, and the clue is an angle. Surveyors, builders and pilots all use angles to find things they cannot measure. We will find angles without using a protractor at all."*

"Notice it, test it, explain why. Today "explain why" has a special meaning: every angle you find needs a REASON."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): The corner of your paper is a right angle, 90°. Draw a line from the corner so that it splits the right angle into two angles, and one of them is 30°. How big is the other angle?	60° (30° + 60° = 90°).
Puzzle 2 (thinking): Two straight lines cross. One of the four angles is 38°. Find the angle next to it. Say which angle fact you used.	142°. Angles on a straight line add up to 180°, and 180° − 38° = 142°.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked." Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Guess the angle. Four angles are drawn on the slide (30°, 90°, 120°, 150°). Use the corner of your paper (90°) and a fold (45°) to estimate each one. Write A, B, C and D guesses, then check.

SAY *"Here are four angles. You are not allowed a protractor. Use the corner of your paper: that is 90°. Fold it in half: that is 45°. Write your guesses on your whiteboard."*

"Which was the hardest to guess? How did you decide?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Angles on a straight line: 180° − 125° = ?	55°
Angles at a point: 150°, 90°, 80° and one more. Find it.	40°
A triangle has angles 90° and 35°. Third angle?	55°
Isosceles triangle, top angle 40°. One base angle?	70°
A parallelogram has a 70° angle. What is the angle next to it?	110°

9–21 min Explore (in pairs, with objects)

SAY *"Draw any triangle and tear off the three corners. What do you think will happen when you put the corners together side by side? Write it down before you try."*

"Now a four-sided shape. Will the four corners fit on a straight line, or round a point?"

DO 1. Draw any triangle. Colour its three corners red, blue and green.

2. Tear off the three corners. Fit them together side by side with the points touching.

3. Draw any four-sided shape and tear off its four corners. Fit them round a point.

4. Say what you notice. Do you think it works for every triangle?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: put the three pointed corners together so that the points touch. Are they flat along the bottom?

Corners do not fit: draw a long straight line first and put the corners along it.

Finished early: draw a triangle with a right angle. Do the corners still fit the same way?

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): Draw any five-sided shape and tear off its five corners. Fit them together. Do they fit on a straight line, or round a point, or neither? Predict first: how many degrees do the five corners make altogether? (Answer: 540°, which is one and a half turns (360° + 180°). A five-sided shape can be cut into 3 triangles from one corner, and 3 × 180° = 540°. A four-sided shape gives 2 triangles, 360°.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Where did the three corners of a triangle fit?	On a straight line: they add up to 180°.
Where did the four corners of a four-sided shape fit?	Round a point: they add up to 360°.
Does it work for every triangle?	Yes, but the best test is to try a very different triangle (fat, thin, right-angled) and see.

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"I want the angle next to 75° on a straight line. A straight line is 180°, so 180 take away 75 is 105. My reason: angles on a straight line add up to 180°."*

"Triangle with 50° and 30°: the three angles add up to 180°, so the third is 180 − 50 − 30 = 100°. My reason: angles in a triangle add up to 180°."

"This triangle is isosceles and the top angle is 50°. The other two angles are equal. 180 − 50 = 130, and 130 ÷ 2 = 65 each. My reason: base angles of an isosceles triangle are equal."

Worked example on the slides	Answer
An angle chase with reasons. Triangle PQR. S is a point on QR. Angle Q is 50°, angle PSQ is 75°, angle R is 30°. Find angle QPS, angle PSR and angle SPR. Give a reason for each.	QPS = 55° (angles in a triangle), PSR = 105° (straight line), SPR = 45° (angles in a triangle). Check: 55° + 45° = 100° = angle QPR.

Rhombus and its diagonal. Rhombus ABCD. Angle DAB is 50°. The diagonal BD is drawn. Find angle ABC, then angle ABD. Give a reason for each.	ABC = 130° (adjacent angles of a rhombus add up to 180°), ABD = 65° (AB = AD, so the base angles are equal: 130° ÷ 2).

SAY *"The key idea today is: one angle, one reason. Find one angle. Say WHY. Use it to find the next one. That is an angle chase. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Measuring or guessing from the picture instead of using an angle fact (drawings are not always to scale).
- Calling the wrong angles equal in an isosceles triangle: the equal angles are opposite the equal sides, which are the base angles, not the angle between the equal sides.
- Giving only one case when one angle of an isosceles triangle is known (the given angle could be the top angle or a base angle).

33–37 min Brain break

DO Angle statues. The tutor calls an angle: 90° is arms in an L, 180° is arms straight out, 45° is arms half-way. Children make it with their arms on camera. Then call 360° (turn all the way round!).

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Maryam Mirzakhani, 1977–2017, Iran and the United States." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: A bouncing ball's path is made of straight lines and angles, and today you find unknown angles on a straight line. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): Angles on a straight line add up to 180°. 180 – 35 – 35 = 110°. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say "Pairs first. Talk before you write."

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

"After question 3 you may choose: carry on, or try Think harder. You can swap at any time."

DO Circulate and ask:

- Ask: “What is your reason for that angle?”
- Ask: “Which angle did you find first? Which angle does it help you find next?”
- Ask: “Do your angles add up to 180° in each triangle? Check.”

IF STUCK Hint 1: “Write down every angle you know first.”

Hint 2: “Look for a straight line, a triangle or a pair of equal sides. Each one gives you an angle fact.”

Hint 3: “Find ONE new angle and give its reason. Then look again: what can you find with it?”

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Angle Chasers (Find all the ways)	First step for everyone: Easy first step: start with a base angle of 10°. The other base angle is also 10°. What is the top angle? Write it as three angles. Then try 20°. (a) An isosceles triangle has all three angles equal to a whole number of TENS of degrees (10°, 20°, 30° and so on). Find ALL the possible triangles. Write each as three angles. (b) How many of your triangles have an angle bigger than 90°? Which one is equilateral? (c) What if the angles are multiples of 15° instead? How many triangles now? What stays the same in your method?
What to look for / model answer	Part 1: eight triangles, written (top, base, base): 160, 10, 10; 140, 20, 20; 120, 30, 30; 100, 40, 40; 80, 50, 50; 60, 60, 60; 40, 70, 70; 20, 80, 80 (every base angle from 10° to 80°; 90° base angles leave 0° at the top, so no triangle). Part 2: four have an angle over 90° (160, 140, 120 and 100); the equilateral one is 60, 60, 60. Part 3: multiples of 15° give five: 150, 15, 15; 120, 30, 30; 90, 45, 45; 60, 60, 60; 30, 75, 75. Method that stays the same: choose a base angle, then top = 180° − 2 × base. Checked by computer for both sets. Look for: a systematic list by base angle and the reason it must stop at 80° (or 75°).

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: In parallelogram ABCD, angle A is 3 times angle B. Find angle A and angle B. Give your reason.	A = 135° and B = 45°. Adjacent angles of a parallelogram add up to 180°, so A and B make 4 equal parts of 45°: B = 45°, A = 135°.

Workbook Think harder 2: An isosceles triangle has an angle of 110°. Find the other two angles. Why is there only one answer?	35° and 35°. 110° cannot be a base angle (two of them would be 220°), so it is the top angle and 180° − 110° = 70° is shared by two equal base angles.
Workbook Think harder 3: In trapezium ABCD, AB is parallel to DC. Angle DAB is 84° and angle ABC is 97°. Find angle ADC and angle BCD.	ADC = 96° and BCD = 83° (angles between the parallel sides add up to 180°). Check: 84° + 97° + 96° + 83° = 360°.
Workbook Think harder 4: ABCD is a square. CDE is an equilateral triangle, and E is outside the square. Find angle ADE and angle DAE.	ADE = 150° (90° + 60°). Triangle ADE is isosceles (AD = DE), so DAE = (180° − 150°) ÷ 2 = 15°.
Workbook Go further 5: ABC is an isosceles triangle with AB = AC and angle BAC = 40°. D is a point on AC with BD = BC. Find angle ABD.	30°. Angles B and C are (180° − 40°) ÷ 2 = 70°. In triangle BCD, BD = BC, so angle BDC = angle C = 70°, which makes angle CBD = 40°. Then ABD = 70° − 40° = 30°.
Workbook Go further 6: ABCD is a square. E is a point inside it so that ABE is an equilateral triangle. Find angle DEC.	150°. AE = AB = AD, so triangle ADE is isosceles with angle DAE = 90° − 60° = 30°, which makes angle AED = 75°. In the same way angle BEC = 75°. Angle AEB = 60°. Round E: 360° − 75° − 75° − 60° = 150°.
Workbook Go further 7: In an isosceles triangle, one angle is 30° bigger than another angle. Find all the possible sets of three angles.	Two sets: 50°, 50°, 80° and 40°, 70°, 70°. (If the top angle is 30° bigger than a base angle: 3 equal parts and 30° make 180°, so base 50°. If a base angle is 30° bigger than the top: top 40°, base 70°.)

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Ravi says an isosceles triangle can have an angle bigger than 90°. Mei says it cannot. Who is right?"*

"Can an isosceles triangle have a right angle? What are the other two angles?"

Discussion question	Listen for
Ravi says an isosceles triangle can have an angle bigger than 90°. Mei says it cannot. Who is right? Can it have a right angle? Find the other two angles each time.	Ravi is right. An isosceles triangle can have one big angle: for example 120°, 30°, 30°. It cannot have two angles of 90° or more, because they would already add up to 180° or more. It can have exactly one right angle: 90°, 45°, 45°. Mei is thinking of the base angles, and she is right about those: the two equal base angles are always less than 90°.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
In triangle PQR, PQ = PR and angle QPR = 44°. Find angle PQR. Write the reason for each step.	68° (angles in a triangle add up to 180°: 180° − 44° = 136°; PQ = PR so the base angles are equal: 136° ÷ 2 = 68°)

SAY *"Today you chased angles with reasons. A detective who can say why is much harder to argue with. Well done, inquirers!"*

"Show me your exit ticket and I will stamp your inquiry journal."

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "Writing the angle on the diagram as soon as you found it got you unstuck."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

Draw a triangle and put the three angle labels 40°, 60° and 80° on the corners. Cut the corners off and fit them together. Now do the same with any four-sided shape. Explain in one sentence why a four-sided shape always gives 360°. (Hint: draw a line from one corner to the opposite corner.)

Answer key (workbook)

Q1 (a) 112° (angles on a straight line: 180° − 68°) (b) 68° (vertically opposite AOC) (c) 112° (vertically opposite BOC) (d) 65° (angles at a point: 360° − 95° − 130° − 70°)

Q2 (a) Parallelogram: B = 108° (adjacent angles add up to 180°), C = 72° (opposite A), D = 108° (opposite B) (b) Rhombus: PQR = 116° (180° − 64°). Triangle PQR is isosceles (PQ = QR), so QPR = (180° − 116°) ÷ 2 = 32° (c) Trapezium: D = 70° (180° − 110°), C = 100° (180° − 80°)

Q3 (a) ADB = 62° (AB = AD, so the base angles are equal) (b) BAD = 56° (180° − 62° − 62°) (c) ADC = 118° (angles on a straight line: 180° − 62°) (d) ACD = 24° (180° − 118° − 38°)

Q4 (a) ADC = 56° (opposite angles of a parallelogram are equal) (b) DEA = 62° (DA = DE, so (180° – 56°) ÷ 2) (c) AEC = 118° (straight line: 180° – 62°) (d) EAB = 62° (angle DAB = 180° – 56° = 124°, then 124° – 62°)

Q5 (a) Two triangles: 50°, 50°, 80° (the 50° angle is a base angle) or 50°, 65°, 65° (50° is the top angle) (b) Only one: 100°, 40°, 40°. The 100° angle must be the top angle, because two angles of 100° would already be 200°.

Story try-it (optional, Maryam Mirzakhani) Angles on a straight line add up to 180°. 180 – 35 – 35 = 110°.

Lesson 9: Pattern Generalisers

Week 5 (first lesson) • Patterns and Number Sense • Workbook: Lesson 9

Objective	Continue growing patterns drawn as figures, say the rule for any figure in words (for example, number of sticks = 3 × figure number + 1) and explain it from the picture, work backwards from a total to find the figure, and solve “two kinds of things” problems with a guess-and-check table and with supposition.
Thinking skills	Looking for a pattern, making a table, describing the rule in words, working backwards, guess and check, supposition, explaining from a picture
Materials	Matchsticks or toothpicks (about 20 per child, 40 if possible), square paper tiles or the square grid from the cut-out pages, mini whiteboard, workbook.
Online set-up	Children have toothpicks and paper squares ready on the desk and build each figure on camera. The tutor shares the slides; pairs work in breakout rooms for the explore and page 1, and hold their tables up to the camera.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *“Today you become pattern generalisers. That is a big word for a simple, powerful idea: if you can see HOW a pattern grows, you can jump straight to the 100th figure without drawing 100 pictures.”*

“Mathematicians and coders do this all the time. By the end of today you will say the rule for any figure in words, and use it backwards too.”

DO Greet each child by name. Show the lesson title slide. Read “Today we will…” aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): Sticks make a row of triangles: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. Build or draw 4 triangles. How many sticks does it use?	9 sticks. Each new triangle shares a side with the one before, so it adds 2 sticks.
Puzzle 2 (thinking): Using the same pattern, how many sticks do 10 triangles need? How many triangles can you make with exactly 41 sticks? Say how you know.	21 sticks for 10 triangles (3 + 9 × 2, or 2 × 10 + 1). 41 sticks make 20 triangles (take away the 1 extra stick, then 40 ÷ 2). Continuing the list also works.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked."

Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Guess My Rule. Call out a number; the tutor puts it through a secret two-step machine and says what comes out. Test your idea on the next number before you say the rule in words. 3 rounds.

SAY *"I have a secret machine. It does two things to every number. You call out a number, I tell you what comes out."*

"When you think you know the rule, do not shout it. Test it on the next number first. Then say it in words."

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
7, 11, 15, 19, __, __	23, 27
1, 4, 9, 16, 25, __	36
2, 6, 18, 54, __	162
The 20th even number	40
4 × 25 + 3	103

9–21 min Explore (in pairs, with objects)

SAY *"If all 8 of us high-five everyone else once, how many high fives is that? Write your guess in the corner of your whiteboard. Is it more or less than 50?"*

DO 1. Guess first: if all 8 of us high-five everyone else once, how many high fives is that?

2. Start small: 2 children, then 3, then 4. Draw each child as a dot and each high five as a line.

3. Make a table: number of children and number of high fives.

4. Look for a pattern. Use it to find the number for 8 children.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: start with 2 children. Then 3. Draw each child as a dot and each high five as a line.

Stuck on the table: when a new child joins, how many NEW high fives does that child make?

Finished early: how many high fives for a class of 30? Is there a quick way without adding 29 numbers?

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): If 12 children each high-five every other child exactly once, how many high fives is that? Find a rule you could use for ANY number of children. (Answer: 66. With n children each high-fives (n – 1) others, but every high five is counted twice, so the number is n × (n – 1) ÷ 2. For 12 children: 12 × 11 ÷ 2 = 66. The table 1, 3, 6, 10, 15, 21 … also works.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many high fives for 5 children?	10. The table goes 1, 3, 6, 10: each new child high-fives everyone already there (add 2, then 3, then 4).
How many for our group of 8?	28 (10 + 5 = 15, + 6 = 21, + 7 = 28).
How many for a class of 30? Is there a quick way?	435. Each of the 30 children high-fives 29 others: 30 × 29 = 870, but that counts every high five twice, so 870 ÷ 2 = 435.

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Squares in a row. Figure 1 has 4 sticks, figure 2 has 7, figure 3 has 10. Each new square adds 3, not 4, because it shares a side with the square before. So I start with 1 stick and add 3 for every square: sticks = 3 × figure number + 1."*

"For figure 25 I do not draw 25 squares. 3 × 25 + 1 = 76. And backwards: 61 sticks. Undo the last step first: 61 − 1 = 60, then 60 ÷ 3 = 20. Figure 20. I check: 3 × 20 + 1 = 61."

"Supposition. 24 vehicles with 70 wheels. Suppose they are ALL motorcycles: 48 wheels. I am 22 wheels short. Each time a motorcycle becomes a car I get 2 more wheels, so 22 ÷ 2 = 11 cars. Check both facts: 11 + 13 = 24 vehicles and 44 + 26 = 70 wheels."

Worked example on the slides	Answer

Matchstick squares in a row: 4, 7, 10 sticks. Figure 25? Which figure uses 61 sticks?. Each new square shares a side, so it adds 3 sticks. Rule in words: sticks = 3 × figure number + 1. Figure 25: 3 × 25 + 1 = 76. Backwards: 61 − 1 = 60, 60 ÷ 3 = 20.	Figure 25 has 76 sticks; 61 sticks is figure 20. Check: 3 × 20 + 1 = 61.
A row of yellow tiles with a grey border: figure 1 has 8 grey tiles, figure 2 has 10, figure 3 has 12. How many grey tiles in figure 12?. Way 1: a top row and a bottom row of (figure + 2) tiles, and 1 tile at each end: 2 × figure number + 6. Way 2: the whole rectangle is 3 rows of (figure + 2) tiles; take away the yellow tiles: 3 × 14 − 12. Both ways: 30.	Figure 12 has 30 grey tiles. Two ways of seeing, one answer.

SAY *"The key idea today is: find the rule, say it in words. See what stays the same and what grows. Say the rule in words: sticks = 3 × figure number + 1. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Doubling: thinking figure 20 has twice as many as figure 10 (62 sticks instead of 61), because the starting stick or the end chairs get counted twice.
- Counting the jumps wrongly: adding 25 lots of 3 to figure 1 for figure 25, instead of 24 lots, or writing the rule as "add 3" without saying what to do with the figure number.
- Supposition: dividing the gap by the value of one thing (22 ÷ 4) instead of by the change for each swap (22 ÷ 2), and not checking the answer against BOTH facts (how many things and the total).

33–37 min Brain break

DO Pattern moves: clap, clap, stamp, clap, clap, stamp … Do 12 moves together. Then: what is the 20th move? The 30th?

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: Al-Karaji, about 953–1029, Baghdad (today Iraq)." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today you look for patterns, say the rule in words and test it with more terms, which is exactly what Al-Karaji did with his cube pattern. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): 1 + 8 + 27 + 64 = 100. 1 + 2 + 3 + 4 = 10 and 10 × 10 = 100. They match. For up to 5: 1 + 2 + 3 + 4 + 5 = 15, so the answer is 15 × 15 = 225 (check: 100 + 125 = 225). Keep the label: His birthplace and the origin of his name are debated by historians. Some of his work survives only through later writers. SKIP IT if you are short of time or the group needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say "Pairs first. Talk before you write."

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

"After question 3 you may choose: carry on, or try Think harder. You can swap at any time."

DO Circulate and ask:

- Ask: "What does each new figure add? Show me where it is in the picture."
- Ask: "Say your rule in words. Does it work for figure 1, 2 AND 3?"
- Ask: "What did you suppose? How much does the total change for each swap?"

IF STUCK Hint 1: "Make a table: figure number in one row, count in the other."

Hint 2: "Colour the part that stays the same in one colour and the part that grows in another."

Hint 3: "To go backwards, undo the steps in reverse order: take away first, then divide."

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Pattern Generalisers (Is it always true? Test and explain)	First step for everyone: Easy first step: fill in the first three totals: 1 = 1, 1 + 3 = 4, 1 + 3 + 5 = ___. Then carry on. (a) Add the first odd numbers: 1, then 1 + 3, then 1 + 3 + 5, then 1 + 3 + 5 + 7. Make a table of the first 6 totals. What do you notice? (b) Is it always true? Use a picture of square dots (add an L-shaped border each time) to explain why. (c) What if you add the first EVEN numbers instead (2, then 2 + 4, then 2 + 4 + 6 ...)? Find a rule in words. Use it to predict the total of the first 20 even numbers, then check it another way.

What to look for / model answer	Part 1: totals 1, 4, 9, 16, 25, 36: the square numbers (1 × 1, 2 × 2, 3 × 3 ...). Part 2: always true (checked for the first 3000 odd numbers): each new odd number is an L-shaped border of dots around the square (3 dots round a 1 by 1, 5 dots round 2 by 2, 7 round 3 by 3), which turns an n by n square into an (n + 1) by (n + 1) square. Part 3: first even numbers add to 2, 6, 12, 20, 30, 42: each is the figure number times the next number (n × (n + 1)). First 20: 20 × 21 = 420. Check: twice the sum of 1 to 20 (210), or "two columns of the 20 by 21 rectangle". Look for: a table, the L-shape picture, and a rule in words, not just examples.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Beads repeat in the order red, red, blue, yellow, yellow, yellow. What colour is the 50th bead? How many yellow beads are in the first 100?	Red (50 = 8 × 6 + 2, so it is the 2nd bead of a group). 49 yellow beads: 16 full groups of 6 give 96 beads and 48 yellow, and beads 97 to 100 are red, red, blue, yellow.
Workbook Think harder 2: 40 square tables in one long row seat 2 × 40 + 2 = 82 people. How many can sit if the tables are put in 8 rows of 5 tables instead?	96 (each row of 5 seats 12, and 8 × 12 = 96). One long row seats only 82. Splitting the tables gives more end seats.
Workbook Think harder 3: A farm has chickens and goats. There are 35 heads and 94 legs. How many goats are there?	12 goats and 23 chickens (all chickens would have 70 legs; 24 legs short; each goat adds 2 legs). Check: 48 + 46 = 94.
Workbook Think harder 4: The pattern 5, 8, 11, 14, ... keeps going up by 3. Is 101 in the pattern? Is 100? Explain how you know, and say which number in the pattern it is if it is there.	101 is the 33rd number (5 + 32 × 3 = 101). 100 is not in the pattern: every number is 2 more than a multiple of 3 (5, 8, 11 ... leave remainder 2 when divided by 3), and 100 leaves remainder 1.
Workbook Go further 5: A dot pattern makes diamonds. Figure 1 has 1 dot. Figure 2 has rows of 1, 3 and 1 dots (5 dots). Figure 3 has rows of 1, 3, 5, 3 and 1 dots (13 dots). How many dots are in figure 6?	61. The counts go 1, 5, 13, 25, 41, 61 (add 4, 8, 12, 16, 20). Or: each diamond is two square numbers of dots, 6 × 6 + 5 × 5 = 36 + 25 = 61.
Workbook Go further 6: A quiz has 25 questions. A correct answer scores 4 marks, 1 mark is taken away for each wrong answer, and a blank scores 0. Mei left 3 questions blank and scored 68. How many did she get correct?	18. She answered 22 questions. All 22 correct would be 88 marks; each wrong answer costs 4 + 1 = 5 marks; 88 − 68 = 20, 20 ÷ 5 = 4 wrong. Check: 72 − 4 = 68.

Workbook Go further 7: Ali writes the counting numbers one after another without spaces: 123456789101112131415... What is the 100th digit he writes?	5. The numbers 1 to 9 use 9 digits. That leaves 91 digits: 45 two-digit numbers (10 to 54) use 90 of them, so the 100th digit is the first digit of 55.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Mei says 10 squares in a row need 31 sticks, so 20 squares need 62. Is she right?"*

"In Q4 and Q5, which did you like better: a guess-and-check table or supposition? Why?"

Discussion question	Listen for
Mei says: "10 squares in a row need 31 sticks, so 20 squares need 62 sticks." Is she right? Explain with a picture.	Mei is not right: 20 squares need 3 × 20 + 1 = 61 sticks. The rule is "3 for each square, plus 1 starting stick". Doubling 31 doubles the starting stick too, so it counts one stick too many. The same thing happens with the end chairs in Q1.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Figure 1 of a pattern has 6 dots. Each figure has 4 more dots than the figure before. (a) How many dots are in figure 20? (b) Which figure has 102 dots?	(a) 82 (6 + 19 × 4, or 4 × 20 + 2) (b) Figure 25 (102 − 2 = 100, 100 ÷ 4 = 25)

SAY *"Today you found how patterns grow, said the rule in words, used it forwards and backwards, and solved two-kinds problems by supposing. That is generalising: one rule for every case."*

"Do the exit ticket on your own, then show me."

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (we will look at it together next lesson, in class)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM

STILL STUCK": it happens in the NEXT lesson, in the open session, at a fixed time: minutes 37 to 49 (page 1 pair work). Visit those children FIRST, in their shared breakout room with the camera on and a parent nearby (face to face: at their table, in the open room), and go through the idea they ticked with one worked example. Before that, in the warm-up (minutes 3 to 9) you may re-teach the most-ticked idea to the whole group. All contact with children stays in the open session. Praise strategy, not speed: "Making a table of the first five figures got you unstuck."

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

A square grid of matchsticks: a 1 by 1 grid uses 4 sticks, a 2 by 2 grid uses 12 sticks. How many sticks for a 3 by 3 grid? A 10 by 10 grid? (24 and 220: a 10 by 10 grid has 11 rows of 10 sticks across and 11 columns of 10 sticks down.)

Answer key (workbook)

Q1 (a) 10, 12; 10 tables: 22 chairs (b) Number of chairs = 2 × number of tables + 2 (c) 24 tables (2 × 24 + 2 = 50) (d) No. 20 tables seat 2 × 20 + 2 = 42, not 44. Doubling counts the 2 end chairs twice.

Q2 (a) 5 sticks: the new house shares a wall with the house before (b) Number of sticks = 5 × number of houses + 1; 20 houses need 101 sticks (c) 39 houses (5 × 39 + 1 = 196 sticks), with 4 sticks left over

Q3 8 adult tickets and 7 child tickets (8 × $12 + 7 × $7 = $96 + $49 = $145). Each adult ticket in place of a child ticket adds $5.

Q4 (a) $6.00; up by 30¢ (b) 14 fifty-cent coins (and 16 twenty-cent coins). $10.20 – $6.00 = $4.20; $4.20 ÷ 30¢ = 14. Check: $7.00 + $3.20 = $10.20.

Q5 (a) 15 correct (all correct would be 100 marks; each wrong answer costs 5 + 2 = 7 marks; 100 – 65 = 35, 35 ÷ 7 = 5 wrong). Check: 75 – 10 = 65. (b) Every score is 100 take away a number of 7s (100, 93, 86, 79, ...). 100 – 80 = 20 is not a multiple of 7, so 80 is impossible.

Story try-it (optional, Al-Karaji) 1 + 8 + 27 + 64 = 100. 1 + 2 + 3 + 4 = 10 and 10 × 10 = 100. They match. For up to 5: 1 + 2 + 3 + 4 + 5 = 15, so the answer is 15 × 15 = 225 (check: 100 + 125 = 225).

Lesson 10: Challenge Day

Week 5 (second lesson) • Mixed Strategies • Workbook: Lesson 10

Objective	Choose a way to start a puzzle (draw a model, list in order, make a table, work backwards, guess and check or suppose, use a logic grid), solve competition-style problems, check every clue, and explain why the chosen start works.
Thinking skills	Choosing a strategy, working backwards with fractions, logic grids, models with units, systematic listing, supposition, checking against every clue, explaining
Materials	Four puzzle cards shown on the slide, or sent as images to each breakout room (backwards, logic, counting, guess and check), mini whiteboards, scrap paper, a 3 by 4 grid drawn on paper, certificates, stamps.
Online set-up	Each breakout pair gets one puzzle card on screen and reports back how they started; logic grids and models are shared on screen and annotated together. Children hold their working up to the camera.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0 min Opening (before the clock starts)

SAY *"Welcome to Challenge Day! There is no new topic today. There are puzzles from every lesson, and the only question that matters is: how do you START?"*

"Clever inquirers do not always know the answer straight away. They have a toolbox of starts, and they check and explain."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

0–3 min Sneak-peek puzzles (private answers)

Puzzle	Listen for
Puzzle 1 (access): Ali, Mei and Ravi wear a red shirt, a blue shirt and a green shirt, one each. Ali is not wearing red or green. Mei is not wearing green. Who wears which colour? Move three cards or draw a small table.	Ali blue, Mei red, Ravi green. Ali can only wear blue. Mei is not green and blue is taken, so Mei wears red. Ravi has green.
Puzzle 2 (thinking): Ali spends half of his money on a book. Then he spends half of the money he has LEFT on lunch. He has $6 left. How much did he have at first? Draw a picture or work from the end.	$24. Work backwards: 6 × 2 = 12 before lunch, 12 × 2 = 24 before the book. A bar cut into 4 equal units, 1 unit = $6, also works.

DO Minutes 0 to 3. SNEAK-PEEK PUZZLES. Show the two puzzles one at a time. Say: "These are ideas we have not learnt yet. Guessing is welcome and so is 'I do not know'. Nobody is marked."

Puzzle 1 is one that any child can try with objects, drawing or counting, so everybody can start well. Puzzle 2 is a thinking puzzle.

Answers are shown PRIVATELY, never on boards in front of the group. Face to face: children write on their own paper and show it only to you as you walk round. Online: everyone holds their paper up to the camera at the same moment, or shows it to you only, on camera. Do not read answers out with names, do not keep a tally, do not rank. Then ask for VOLUNTEERS to say how they thought (put a hand up or use a reaction); never choose a child because you saw a wrong answer. Never record scores.

Choosing a start: every child has the same neutral line on workbook page 1: "After question 3 you may choose: carry on, or try Think harder." Invite the WHOLE group, never one child: "when you have finished Start here you can choose Think harder, Go further or the Investigation." Never single out a child by name and never announce who is on which page. Pairs are not split unless BOTH children agree. Any child may swap at any time.

3–9 min Warm-up

DO Minutes 3 to 9: the warm-up game below (one or two rounds only) and then the quick questions. If you are short of time, ask only the first three quick questions.

DO Mix-up relay. One quick question from each earlier lesson. Read a card; a volunteer answers and invites the next volunteer. You may say pass once.

SAY *"Mix-up relay: I read a card from one of our earlier lessons and say a name. That child answers, then names the next child. A wrong answer gets a friendly "try again", never a groan. You may say "pass" once. The child who just answered may choose a green card (a quick one) or an orange card (a think-about-it one). The tutor keeps a private list so that every child gets a turn before anyone has a second, and asks for volunteers first; nobody is called on by name if they would rather not."*

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
What is 3/4 of 48?	36
Area of a triangle with base 10 cm and height 6 cm	30 cm²
A tap fills 6 litres in 2 minutes. How long to fill 45 litres?	15 minutes
Two angles on a straight line are 65° and ___	115°
Next in the pattern: 5, 8, 11, 14, __	17

9–21 min Explore (in pairs, with objects)

SAY *"Look at the four puzzle cards. Which one do you think will be the hardest? Write your guess in the corner of your whiteboard and we will see at the end."*

DO 1. Pick 2 of the 4 puzzle cards. Work in pairs.

2. Write your answer, and write HOW you started (draw, list, table, work backwards, guess and check).

3. Check your answer against every clue.

4. Be ready to teach one card to the group in two sentences.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck: "How could you start? Draw it, list it, make a table, work backwards, or guess and check?"

Backwards puzzle: "Write the steps going forwards first. Then undo the LAST step first."

Logic puzzle: "Find a clue that tells you something for sure. Put a cross where it cannot be."

Finished early: make a puzzle card for the next pair.

DO Go further what-if (on the second explore slide, for anyone who wants more, so nobody waits): Card A has a twin: a number is TRIPLED, then 4 is taken away, then the answer is divided by 2, and the result is 7. What was the number? Then make up your own card of this kind whose answer is 10. (Answer: 6. Work backwards: 7 × 2 = 14, 14 + 4 = 18, 18 ÷ 3 = 6. Check: 6 × 3 = 18, 18 − 4 = 14, 14 ÷ 2 = 7. Any new card is fine if its working backwards gives a whole number and it checks when run forwards.)

At about minute 19, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Card A: a number is doubled, 7 is added, the answer is divided by 3, giving 5. What was the number?	4 (work backwards: 5 × 3 = 15, 15 − 7 = 8, 8 ÷ 2 = 4).
Card B: Ali, Ben and Mei have a cat, a dog and a fish. Ali has no dog. Ben has no cat and no dog. Who has what?	Ben has the fish, Ali the cat, Mei the dog.
Card D: two numbers add to 20 and differ by 6. What are they?	13 and 7 (guess and check, or take away the 6 and share: 14 ÷ 2 = 7, then 7 + 6 = 13).

21–33 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"A number is doubled, then 7 is added, then divided by 3, and the answer is 5. I work backwards and undo the last step first: 5 × 3 = 15, then 15 − 7 = 8, then 8 ÷ 2 = 4. I check forwards: 4 × 2 = 8, 8 + 7 = 15, 15 ÷ 3 = 5."*

"A logic grid. Ravi does not play swimming. Siti plays neither football nor badminton, so Siti swims for sure. A tick means crosses in the rest of that row AND that column. Ben does not play football, so Ben badminton and Ravi football. I check every clue."

"How many squares are in a 2 by 3 grid? If I count only the small ones I get 6, but there are bigger squares. I count by size: 1 by 1: 6. 2 by 2: 2. Total 8. Counting by size means I do not miss any."

Worked example on the slides	Answer
Working backwards: spent 1/4, then $10, then half of the rest. $13 left.. Draw it: start, then each step. Work back from the end. $13 left is half of the rest, so the rest was $26. Before lunch: 26 + 10 = 36. $36 is 3/4 of the money, so 1/4 is $12 and all of it is $48.	$48. Check: 48 − 12 = 36, 36 − 10 = 26, 26 ÷ 2 = 13 ✓
Logic grid: Ravi, Siti and Ben play badminton, swimming and football. Ravi does not swim. Siti does not play football or badminton. Ben does not play football.. Siti: not football, not badminton, so swimming for sure. A tick means crosses in the rest of its row and column. Ben is not football, so badminton; Ravi is football.	Ravi football, Siti swimming, Ben badminton. Check all three clues.

SAY *"The key idea today is: choose a start, check every clue. Draw it, list it, make a table, work backwards, guess and check or suppose, use a logic grid. Then explain why your start works. Say it with me."*

DO Show the key idea slide at minutes 32 to 33. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Working backwards in the wrong order or using the wrong operation, such as taking away instead of adding back (7 + 5 = 12, then doubling for the half that was spent, 24), or halving instead of doubling.
- Stopping when one clue works instead of checking every clue, and in logic puzzles guessing instead of using a clue that tells you something for sure.
- Counting only the smallest squares or rectangles (12 in a 3 by 4 grid) and missing the bigger ones made of several small squares; not counting in order by size.

33–37 min Brain break

DO Puzzle stretch: stand up and show a 3 by 3 grid with your fingers. How many squares can you find on it? Count by size and tell your neighbour.

Water. Toilet break for anyone who needs it. Keep it to four minutes.

DO Story break (optional, 2 minutes): after the stretch and a drink, say "Meet a mathematician: George Polya, 1887–1985, Hungary, Switzerland and the United States." Read the story on the slide aloud, slowly (about one minute), or let a volunteer read it. Link to today: Today is Challenge Day, and Polya's ideas such as drawing a picture, working backwards and checking are the tools you choose from. Then offer the Try it as an OPTIONAL question: "Have a go in your head or on paper, or just listen. Pass is always fine." Volunteers only, nobody is marked. ANSWER (for you; show it only after children have had a go): Undo the steps: 20 − 6 = 14, then 14 divided by 2 = 7. Check: 7 × 2 = 14, 14 + 6 = 20. The number was 7. SKIP IT if you are short of time or the group

needs the full break: say "we will skip the story today" and go straight to the workbook at minute 37; nothing else depends on it.

37–72 min Workbook (page 1 in pairs, check, teaching, page 2 alone)

DO Minute 37: give out the workbook. Say “Pairs first. Talk before you write.”

Minutes 37 to 49: pairs work on page 1. Minutes 49 to 53: check page 1 together (four minutes, so every part can be explained). Minutes 53 to 58: the short teaching slides for the page 2 question types. Minutes 58 to 72: each child works alone on page 2; it is a menu, not a wall, so nobody has to finish, and minutes 62 to 72 are protected open think time for Think harder, Go further or the Investigation. Minutes 72 to 76: check page 2 together.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

“After question 3 you may choose: carry on, or try Think harder. You can swap at any time.”

DO Circulate and ask:

- Ask: "How did you start? Is there another start that would work?"
- Ask: "Did you check your answer against EVERY clue?"
- Ask: "What would you try first if the numbers were much bigger?"

IF STUCK Hint 1: “Draw a bar or a table. Put what you know in it.”

Hint 2: “Which clue tells you something for sure? Start there.”

Hint 3: “Undo the last step first. Fractions: if half was spent, half is left.”

DO Quick finishers: offer the Go further page or the Investigation. Everyone has the same neutral line on page 1. Invite the whole group ("when you have finished Start here you can choose Think harder, Go further or the Investigation"); never single out a child, never announce who is on which page, and do not split a pair unless both children agree.

Investigation	Detail
Investigation (printed on the Investigation and reflection page every child gets): Challenge Day (Find the pattern)	First step for everyone: Easy first step: with 1 step there is 1 way. With 2 steps there are 2 ways (1 + 1, or 2). Draw the ways for 3 steps first. (a) Ali climbs a staircase and takes either 1 step or 2 steps at a time. How many different ways can he climb a staircase with 1 step? 2 steps? 3 steps? 4 steps? List every way for 4 steps (for example 1+1+2 and 2+1+1 are different). (b) Make a table up to 6 steps. What pattern do you see? Explain why it works (hint: think about the LAST jump Ali makes). (c) Use your pattern to predict the number of ways for 8 steps. What if Ali could also take 3 steps? Try 3 and 4 steps again.

What to look for / model answer	Part 1: 1 step: 1 way; 2 steps: 2 ways (1+1, 2); 3 steps: 3 ways (1+1+1, 1+2, 2+1); 4 steps: 5 ways (1+1+1+1, 1+1+2, 1+2+1, 2+1+1, 2+2). Part 2: 1, 2, 3, 5, 8, 13: each is the sum of the two before it. Reason: the last jump is 1 step (so Ali had climbed n − 1 steps before) or 2 steps (n − 2 steps before), so the ways for n steps = ways for n − 1 + ways for n − 2. Part 3: 8 steps gives 34 ways (1, 2, 3, 5, 8, 13, 21, 34; checked by brute force). With 3-steps allowed: 3 steps has 4 ways and 4 steps has 7 (each is the sum of the three before). Look for: a complete, ordered list, the table, and the last-jump reasoning.

Think harder and Go further questions

Question	Answer
Workbook Think harder 1: Siti spent 2/5 of her money, then 1/3 of what was left. She had $24 left at the end. How much did she start with?	$60. Work backwards: $24 is 2/3 of what was left, so that was $36; $36 is 3/5 of her money, so 1/5 is $12 and she started with $60. Check: 60 − 24 = 36, 36 − 12 = 24.
Workbook Think harder 2: How many 3-digit numbers have digits that add up to 5? List them all in order.	15 numbers: 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500.
Workbook Think harder 3: The sum of two consecutive numbers (numbers that are next to each other, like 40 and 41) is 145. What are the two numbers?	72 and 73 (145 is 1 more than twice 72).
Workbook Think harder 4: A snail climbs 3 m up a 10 m wall each day, and slips down 2 m each night. On which day does it reach the top?	Day 8. After 7 days and nights it is at 7 m. On day 8 it climbs 3 m to reach 10 m before it can slip.
Workbook Go further 5: A water tank is 1/3 full. After 20 litres are added it is 1/2 full. Then 1/4 of the water in the tank is let out. How many litres of water are in the tank now? What fraction of the tank is full?	45 litres, which is 3/8 of the tank. The 20 litres are 1/2 − 1/3 = 1/6 of the tank, so the full tank holds 120 litres. Half full is 60 litres; letting out 1/4 of it (15 litres) leaves 45 litres, and 45 ÷ 120 = 3/8.
Workbook Go further 6: 25 coins of 50¢, 10¢ and 20¢ are worth $5.30 altogether. There are twice as many 10¢ coins as 50¢ coins. How many coins of each kind are there?	3 fifty-cent coins, 6 ten-cent coins and 16 twenty-cent coins. Suppose all 25 coins were 20¢: that is $5.00. Every 50¢ coin comes with two 10¢ coins; this group of 3 coins is worth 70¢, and replacing three 20¢ coins (60¢) with it adds 10¢. The extra is 30¢, so there are 3 groups. Check: 150¢ + 60¢ + 320¢ = 530¢ and 3 + 6 + 16 = 25.

Workbook Go further 7: Ali, Ben, Mei and Siti sit in a row of 4 chairs. Mei is not on either end. Ben does not sit next to Mei. Siti sits somewhere to the left of Ali. Ben is not on the left end. In what order do they sit, from left to right?	Siti, Mei, Ali, Ben. Mei is in chair 2 or 3. If Mei is in chair 3, the only chair not next to her is chair 1, but Ben is not on the left end, so that fails. So Mei is in chair 2, the only chair not next to her is chair 4, so Ben is in chair 4. Siti and Ali are in chairs 1 and 3 and Siti is on the left: Siti 1, Ali 3.

These are the questions on the Think harder and Go further pages in the workbook. More are in the separate Question Bank.

76–82 min Talk time

DO Talk-time rules: say "You may say pass." Ask for VOLUNTEERS first, then invite others only if they want to. No cold-calling by name, no public tallies or rankings, and no "who got it right?" hands.

Bring the group back to a circle (or camera on, mics on). Pairs talk for 2 minutes first.

SAY *"Pick one puzzle and tell your partner in two sentences. You can say: I started by ____. I know it is right because ____. You may pass."*

"Which start do you think was the most useful today? When would you NOT use it?"

Discussion question	Listen for
Pick one puzzle and teach it in two sentences: how did you START and how do you know your answer is right? Could another start also work?	Open. Praise reasons, not just answers. Good answers say what the start does: a model shows how the parts relate, working backwards undoes steps in reverse, a logic grid keeps clues tidy, listing by size makes sure nothing is missed. Different starts can both work.

82–90 min Exit ticket, reflection and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Ravi spends 1/2 of his money on a ticket, then $5 on a snack, and has $7 left. How much money did he start with? Say which start you used.	$24. Work backwards: 7 + 5 = 12, and 12 is half of his money, so 12 × 2 = 24. Check: 24 ÷ 2 = 12, 12 − 5 = 7.

SAY *"Today you used a toolbox of starts and checked every clue. That is how problem solvers work, and it is the end of our Math Inquirers program."*

"Do the exit ticket on your own, then show me. Then it is certificate time!"

DO Minutes 86 to 88. REFLECTION, 2 minutes, never skipped. Show the exit slip and reflection slide. Children fill in the small box at the bottom of the Investigation and reflection page: "Something I did well", "The hardest part was...", "What helped me" with tick boxes that include "I asked for help" and "I am still stuck (my parent can ask my tutor about it)", and "Next time I will try ...". Say: "Pass is always OK. You can write one word, tick a box, or write pass."

It is never collected or shown in public. You may read it afterwards to plan the next lesson. Never read a child's reflection aloud; only a child who volunteers may share. FOLLOW-UP FOR "I AM STILL STUCK": Lesson 10 has no next lesson. At the goodbye (88 to 90) tell the whole group: "if something is still tricky, ask your parent to tell me through the usual parent channel and I will send a worked example to your parent." All contact with children stays in the open session. Praise strategy, not speed: "You tried a different way to start when the first one did not work. That is what we practised all holiday." This is the last lesson, so a volunteer may say what they would tell a friend starting the program.

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide (private to you, never shown to children). If a child is marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for anyone who wants more

Write your own logic puzzle with four people and four things. Test it on a friend: does it have exactly ONE answer? Which clue could you take away without ruining it?

Answer key (workbook)

Q1 Ali: chicken rice. Ben: laksa. Mei: satay. Siti: roti prata. (Clue 2 says a girl ordered the roti prata, so it is Mei or Siti, and clue 4 rules out Mei: roti prata is Siti's. Ali is then chicken rice by clue 1. Ben is not satay, so laksa, and Mei has satay.)

Q2 $63. Work backwards: 15 × 2 = 30 before the gift; 30 + 12 = 42 before lunch; 42 is 2/3 of the money, so 1/3 is 21 and the start is 63. Check: 63 − 21 = 42, 42 − 12 = 30, 30 ÷ 2 = 15 left.

Q3 Mei $60, Ravi $55. Model: Mei is 4 units; Ravi is 3 units and $10. 7 units + $10 = $115, so 7 units = $105 and 1 unit = $15. Mei 4 × 15 = $60, Ravi 3 × 15 + 10 = $55. Check: 60 + 55 = 115 and 3/4 of 60 = 45, 45 + 10 = 55.

Q4 (a) 12 squares of size 1 by 1 (b) 6 squares of size 2 by 2 and 2 squares of size 3 by 3 (c) 12 + 6 + 2 = 20 squares. (There are none bigger: the grid is only 3 rows tall.)

Q5 Ben broke the vase. Try each child: if Ali did it, Ali and Ben both lie (2 liars); if Mei did it, Ali and Mei both lie (2 liars); if Ben did it, only Ben lies (1 liar). Only Ben fits.

Q6 (a) In 7 years: Mum will be 48 and Mei 16, and 3 × 16 = 48. (Table: after 1 year 42 and 10; 2 years 43 and 11; ... 7 years 48 and 16.) Quick way: Mum is always 32 years older. When she is 3 times as old, the 32 years are 2 times Mei's age, so Mei is 16, which is 7 years from now. (b) Open: for example "I made a table of their ages", "I used the age gap, which never changes" or "I guessed and checked".

Story try-it (optional, George Polya) Undo the steps: 20 − 6 = 14, then 14 divided by 2 = 7. Check: 7 × 2 = 14, 14 + 6 = 20. The number was 7.

Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

Student	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Notes

Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

Tutor checklist

Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

During the lesson

- Circulate and ask questions rather than giving answers.
- Contact parents only through the approved channels. All contact with children stays in the open session: invitations are made to the whole group, and check-ins happen in the shared room with a parent nearby.
- Do not take photos or recordings unless the Company has approved it and parents have consented.
- If you have a concern about a child's safety or welfare, tell the Company straight away.

After the lesson

- Stamp inquiry journals and fill in the tracker.
- Collect worksheets and answer keys (face to face) and keep them secure.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

A 10-lesson holiday maths program for children whose school year covers Primary 5 topics

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are live online in small groups, hands-on, and built around thinking, not speed.

What your child will do

- Lesson 1, Million Puzzles: numbers to 10 million
- Lesson 2, Fraction Feats: fractions: four operations
- Lesson 3, Before and After: fraction word problems
- Lesson 4, Triangle Areas: area of triangle
- Lesson 5, Volume Vault: volume
- Lesson 6, Rate Racers: rate
- Lesson 7, Percentage Puzzles: percentage
- Lesson 8, Angle Chasers: angles, triangles and quadrilaterals
- Lesson 9, Pattern Generalisers: patterns and number sense
- Lesson 10, Challenge Day: mixed strategies

How you can help

- Ask “How do you know?” when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Print the workbook and cut-outs before the lesson, and keep scissors, a ruler and a pencil ready.
- Lesson 5: 24 to 36 small cubes (dice, Lego, sugar cubes) or squared paper. Lesson 9: 20 matchsticks or toothpicks (or squared paper).

Practical details

Dates	[start date] to [end date], two lessons a week
Time	[day] and [day], [start time] to [end time]
Where	Live online video class (link sent after registration)
Group size	[number] children
Fee	S$500 introductory (regular S$650), workbook included, payable in 3 instalments
Contact	[name, phone or email]

Math Inquirers Holiday Program is an independent enrichment program. It builds the thinking skills of problem solving but is not affiliated with, run by or endorsed by the Ministry of Education or any school.